

***Angelica gigas*: Signature Compounds, In Vivo Anticancer, Analgesic, Neuroprotective and Other Activities, and the Clinical Translation Challenges**

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Abstract: *Angelica gigas* Nakai (AGN) root is a medicinal herbal widely used in traditional medicine in Korea. AGN root ethanolic extract dietary supplements are marketed in the United States for memory health and pain management. We comprehensively reviewed the anticancer, analgesic, pro-memory and other bio-activities of AGN extract and its signature phytochemicals decursin, decursinol angelate, and decursinol a decade ago in 2012 and updated their anticancer activities in 2015. In the last decade, significant progress has been made for understanding the pharmacokinetics (PK) and metabolism of these compounds in

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animal models and single dose human PK studies have been published by us and others. In addition to increased knowledge of the known bioactivities, new bioactivities with potential novel health benefits have been reported in animal models of cerebral ischemia/stroke, anxiety, sleep disorder, epilepsy, inflammatory bowel disease, sepsis, metabolic disorders, osteoporosis, osteoarthritis, and even male infertility. Herein, we will update PK and metabolism of pyranocoumarins, review *in vivo* bioactivities from animal models and human studies, and critically appraise the relevant active compounds, the cellular and molecular pharmacodynamic targets, and pertinent mechanisms of action. Knowledge gaps include whether human pyranocoumarin PK metrics are AGN dose dependent and subjected to metabolic ceiling, or metabolic adaptation after repeated use. Critical clinical translation challenges include sourcing of AGN extracts, product consistency and quality control, and AGN dose optimization for different health conditions and disease indications. Future research directions are articulated to fill knowledge gaps and address these challenges.

Keywords: *Angelica gigas*; *In Vivo* Bioactivities; Pyranocoumarins; Decursin; Decursinol; Review.

Introduction

We comprehensively reviewed the anticancer and other bio-activities of *Angelica gigas* Nakai (AGN, commonly known as Korean Angelica or Korean Dang Gui) and its phytochemicals, including pyranocoumarin compounds decursin (D), decursinal angelate (DA) and decursinol (DOH) (See structures in Fig. 1A) a decade ago (Zhang *et al.*, 2012a) and updated their anticancer activities in 2015 (Lu *et al.*, 2015). In the last decade, significant progress has been made for understanding the pharmacokinetics (PK) and metabolism of these signature compounds in animal models. Two human single dose PK studies have been published by us (Zhang *et al.*, 2015a) and a Korean group (Kim *et al.*, 2018b). In addition to gain of knowledge for the known bioactivities, potential novel health benefits have been reported in animal models, including preventing or treating cerebral ischemia/stroke, anxiety, sleep disorder, epilepsy, inflammatory bowel disease, sepsis, metabolic disorders (syndrome), osteoporosis, osteoarthritis, and male infertility. In this paper, we will update the PK and metabolism of pyranocoumarins, review and update *in vivo* bioactivities from animal models and human studies, and critically appraise the relevant active compounds and the cellular and molecular pharmacodynamic (PD) targets. In addition to identifying knowledge gaps, we will discuss critical clinical translation challenges and suggest future research directions. Readers are referred to our comprehensive review paper (Zhang *et al.*, 2012a) and the original references therein for historical information and specific details.

Major Oriental Medicinal *Angelica* Species

Three major species of *Angelicas* are used in herbal medicine in China, Korea, and Japan: *Angelica sinensis* (Chinese Dang gui, or Donquai), *Angelica gigas* Nakai (Korean) and *Angelica acutiloba* (Japanese). Molecule-based classification in addition to morphology is very crucial to detect and prevent adulteration of the dried roots of different *Angelica*

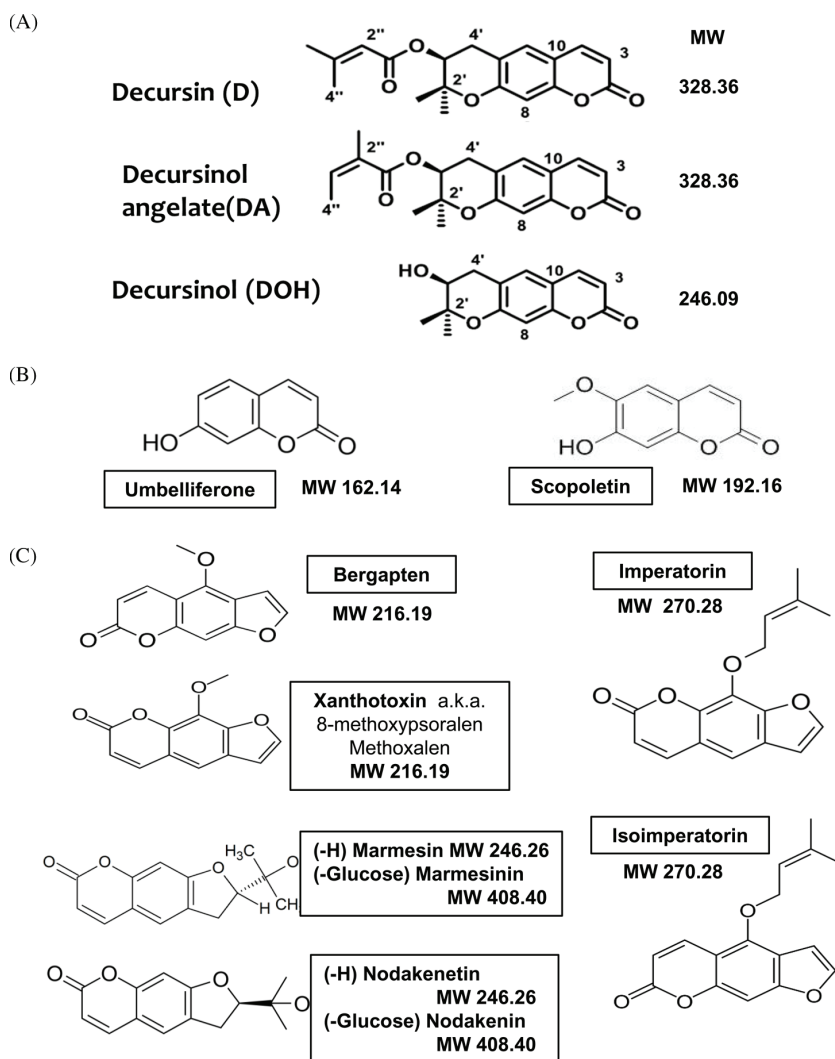


Figure 1. Chemical structures of *Angelica gigas* Nakai (AGN) root phytochemicals in alcoholic extracts. (A) Signature pyranocoumarins decursin (D), decursinol angelate (DA) and decursinol (DOH). (B) Simple coumarins (umbelliferone, scopoletin). (C) Furanocoumarins and their glucosides.

species available in the herbal markets. Genetic differences among *A. sinensis* (Chinese), *A. acutiloba* (Japanese) and AGN (Korean) have been documented, including 5S-rRNA spacers (Zhao *et al.*, 2003). They are phylogenetically divided into two clusters: the first group consisting of *A. sinensis* and the second group consisting of AGN and *A. acutiloba* (Zhao *et al.*, 2003). Chemically, the contents of ferulic acid and Z-ligustilide in the root of *A. sinensis* were much higher than those of *A. acutiloba* and AGN (Zhao *et al.*, 2003). Others have profiled the volatile components in *Angelica* species by gas chromatographic/mass spectrometry (GC/MS) (Kim *et al.*, 2006a; 2006b). GC/MS profiling was used as a pattern recognition method for the quality control of AGN (Piao *et al.*, 2007) with a

principal component analysis (PCA) subsequently applied to 10 common peaks selected from each chromatogram. Profiling chemical components of AGN from different geographical origins using ^1H NMR and UPLC-MS analyses identified pyranocoumarins decursin (D), decursinal angelate (DA) (See structures in Fig. 1A), and furanocoumarins nodakenin, marmesin (See structures in Fig. 1C) and 7-hydroxy-6-(2R-hydroxy-3-methylbut-3-ethyl) coumarin as major phytochemicals that contributed to the discrimination factor (Kim *et al.*, 2011). Taken together, D and DA are a robust identifying chemical characteristic of AGN from other major *Angelica* species. Given the phytochemical differences, it is not a surprise that these herbals have different medicinal properties, and each has been subjected to intense investigations and herbal medicinal development in the respective countries.

AGN Nutraceutical Preparations and Signature Phytochemicals

AGN roots are traditionally used to treat anemia, cold, pain and other diseases as folk medicines and known to Korean herbalists as “female ginseng” (Zhang *et al.*, 2012a). However, these traditional medicinal properties were mostly based on using boiling water to extract the active ingredients. Changing the extraction solvents can lead to the recovery of different chemicals and therefore novel medicinal and pharmacological properties. In fact, many pyranocoumarin compounds have been identified from the alcoholic extracts of AGN roots, which could not be extracted by water. Chemical compositions of roots of the *Angelica* genus have been extensively studied [see review (Zhang *et al.*, 2012a)]. The different types of AGN root preparations and extracts and their respective signature phytochemicals are schematically illustrated in Fig. 2. Double line boxed entities have been evaluated for *in vivo* bioactivity.

Alcoholic Extracts

The majority of reported studies have used alcoholic extracts or compounds further purified from them. Pyranocoumarins are the major chemicals identified in the ethanolic or methanolic extracts of AGN root. Decursin (D) (Fig. 1A), the most abundant pyranocoumarin, was first isolated from the root of *Angelica decursiva* FR and SAV and later from AGN in the 1960s. Decursinol angelate (DA) (Fig. 1A) is a structural isomer of D on the side chain and about 50–60% as abundant as D in the root of AGN. The pyranocoumarin core decursinol (DOH) (Fig. 1A) can be detected with a much lower abundance, if any. Additional structurally related chemicals include 4''-hydroxytigloyldecursinol, 4''-hydroxydecursin, (2''S,3''S)-epoxyangeloyldecursinol and (2''R,3''R)-epoxyangeloyldecursinol (Zhang *et al.*, 2012a). The major alcohol-extractable pyranocoumarins (D/DA) may constitute 3.0–8.2% of dried AGN root or as high as 60% of the ethanol extract, depending on the extraction protocols and sourcing of AGN roots.

Beside these pyranocoumarins, other compounds included simple coumarins (Fig. 1B) such as umbelliferone (Konoshima *et al.*, 1968; Jung *et al.*, 1991), and scopoletin (Park *et al.*, 2011) and furanocoumarins (Fig. 1C), for example, bergapten (Park *et al.*, 2011), xanthotoxin (Park *et al.*, 2011), imperatorin (Park *et al.*, 2011), isoimperatorin (Reddy *et al.*, 2008;

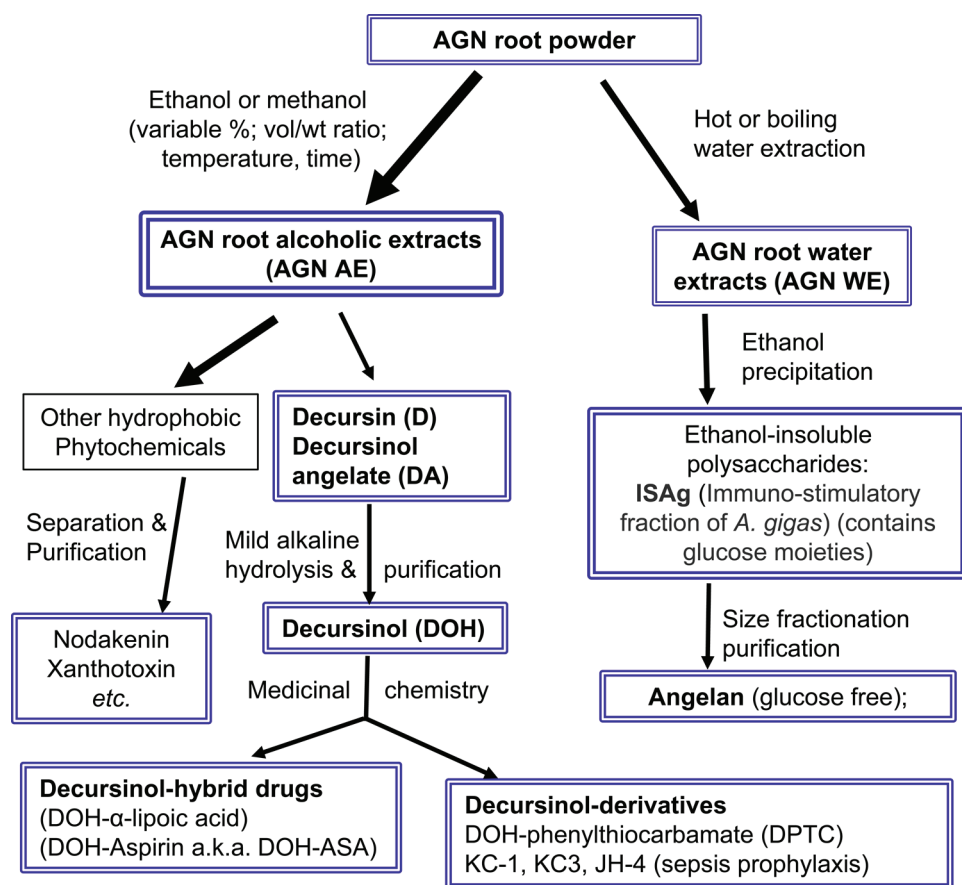


Figure 2. Reported *Angelica gigas* Nakai (AGN) root extract preparations, signature phytochemicals and medicinal chemistry modifications of decursinol (DOH-hybrid drugs and DOH derivatives). Entities in double-lined boxes have been tested for at least one bioactivity in an animal model.

Yoo *et al.*, 2008), marmesin (Kim *et al.*, 2006b; Reddy *et al.*, 2008; Yoo *et al.*, 2008) and its glucoside marmesinin (Kang and Kim, 2007), and nodakenetin (Konoshima *et al.*, 1968; Jung *et al.*, 1991) and its glucoside nodakenin (Jung *et al.*, 1991; Yoo *et al.*, 2008). See our earlier review for additional chemical structures and the botanical biosynthetic pathways for DOH core and the more complex pyranocoumarins (Zhang *et al.*, 2012a).

There were many variables in the process of AGN root extract preparation. The alcoholic extracts were prepared with either ethanol or methanol, with their % varying from 30% to absolute. The ratios of root solid to solvent volume (mass/vol.) and the number of solvent changes varied by different labs. The durations of each extraction could vary, and the extraction temperature could range from room temperature to elevated thermal extraction. The physical state of root could be dried slices vs. powdered finer particles. The extent of agitation during each extraction could affect yield as well.

The resulting liquid would be filtered through filter papers for analytical lab preparations or cheese cloths or other sizing materials for large or industrial scale production to rid of particulates and subsequently rendered “solvent-free”. These diverse extraction conditions led to considerable variability of D/DA content and other components. The ethanol extract coded INM-176 (for Institute of Natural Medicine) has been exclusively used in the marketed dietary supplements in the USA, such as in CognI.Q (Quality of Life Laboratories, Purchase, NY).

Water Extracts

In general, ground root powder was soaked and agitated in a sufficient volume of warmed or boiling water and filtered to get rid of particulates (See Fig. 2 schema). The ethanol-precipitable (insoluble) components of the aqueous extracts of AGN root were used either directly (such as ISAg, short for immunostimulatory component of *A. gigas*) (Kim *et al.*, 2018a; Lee *et al.*, 2019a) or further sub-fractionated into more refined polysaccharides (e.g., angelan) (Han *et al.*, 1998; Han *et al.*, 2006; Kim *et al.*, 2008) for studies of immune regulatory activities and cancer control in syngeneic mouse models. These preparations contain predominantly polysaccharides and 9–17% proteins. One AGN water extract (condensed volume without further organic solvent partitioning) was tested for cerebral ischemia stroke protection in rats (Oh *et al.*, 2015). The composition of this preparation would be expected to be more complex than angelan or ISAg. All these water extracts should contain little or no pyranocoumarins or hydrophobic secondary metabolites.

Whole Root Powders

Finely ground powders of AGN root have been tested in ovariectomized (OVX) rats against ovarian hormone-deficit osteoporosis (Choi *et al.*, 2012). Nanoparticle formulations of AGN root powder was also tested for memory benefit (Piao *et al.*, 2015). These herbal preparations would be expected to contain both the primary fibrous structural components (cellulose) as well as secondary phytochemical metabolites, including D/DA.

Decursinol Derivatives and Hybrid Drugs

Medicinal chemistry modifications of DOH have been reported by us and others aiming for improving drug-likeness or efficacy (Fig. 3). We synthesized decursinol phenylthiocarbamate (DPTC) to preserve *in vivo* androgen receptor (AR) targeting activity of D and DA (Zhang *et al.*, 2012b). While the side chain modification preserved *in vitro* and *in vivo* AR suppressing activity, DPTC was poorly bioavailable due to extremely low solubility (Zhang *et al.*, 2012b). Others (Lee *et al.*, 2012; Yan *et al.*, 2013) reported hybrid drugs between DOH and alpha-lipoic acid (DOH-ALA) or aspirin (DOH-aspirin or DOH-ASA for acetyl salicylic acid) against ischemia reperfusion stroke models. It is remarkable that the hybrid drugs showed demonstrable prophylaxis efficacy when DOH or the respective counterpart drug failed to protect

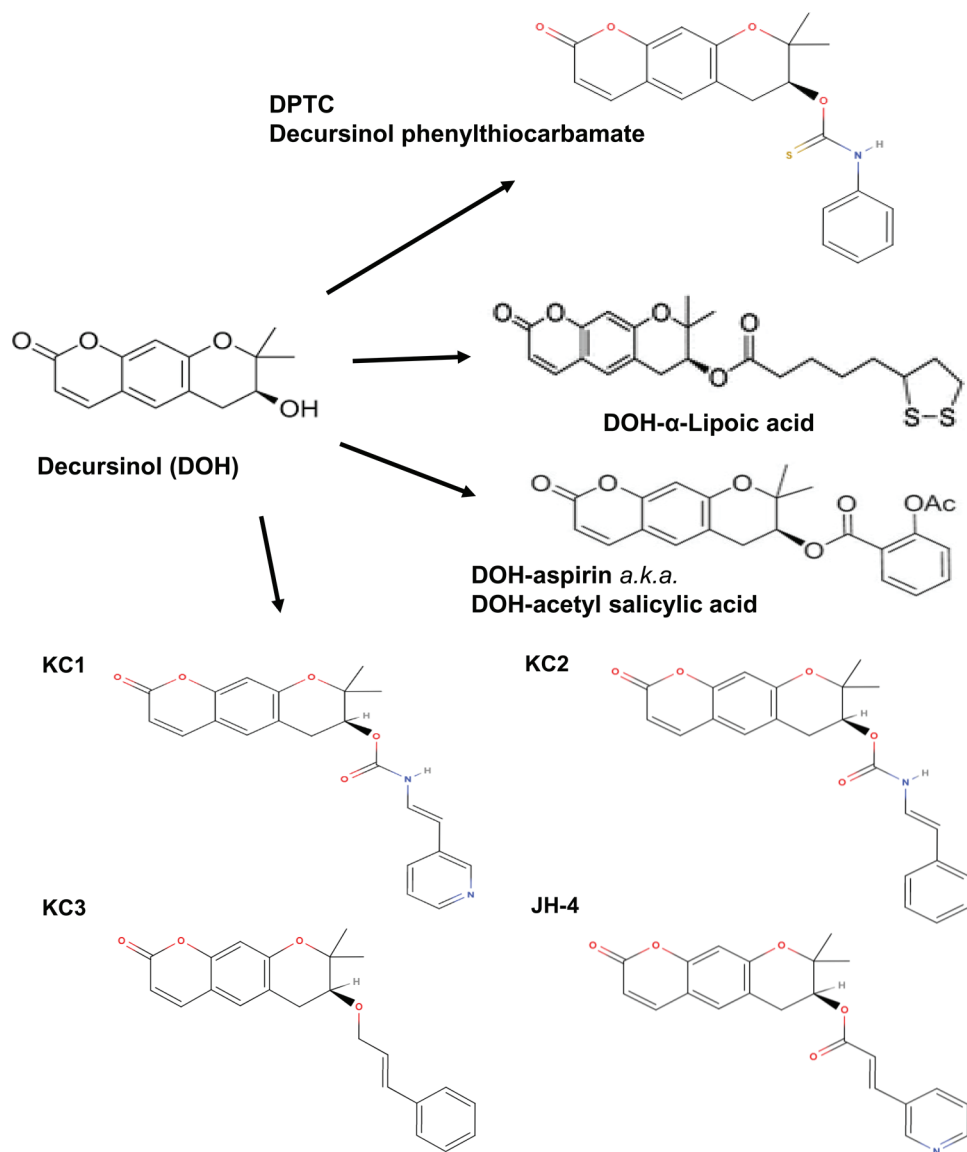


Figure 3. Reported medicinal chemistry modifications of decursinol. DPTC was examined as an androgen receptor signaling inhibitor. Decursinol-hybrid drugs were tested for ischemia reperfusion stroke prevention. Decursinol analogs KC1-3 and JH-4 were tested for prophylaxis of sepsis.

against ischemia stroke at the same dosage (20 mg/kg; in spite of bias against the hybrids on molar dosage basis). In these hybrid drug scenarios, DOH might have acted as a neuron/brain drug carrier to enhance delivery of lipoic acid and aspirin payloads or the hybrids represented novel drug entities distinct from their components. Lee *et al.* synthesized and reported 4 DOH

analogs, 3 of which (KC1, KC3, JH4) had remarkably improved potency on prophylaxis of sepsis models (ED₅₀–0.6 mg/kg vs. ~10 mg/kg for DOH) (Lee *et al.*, 2019b; 2019c).

Pharmacokinetics and Metabolism of D, DA and DOH

PK Studies in Rodent Models

Although D and DA have been selected for evaluation in most cancer cell culture models and many other cell types, rodent model studies have shown a rapid and extensive conversion of D and DA to DOH as summarized below and was confirmed in our human PK study (Zhang *et al.*, 2015a) (Fig. 4). The PK knowledge casts grave doubts over the *in vivo*

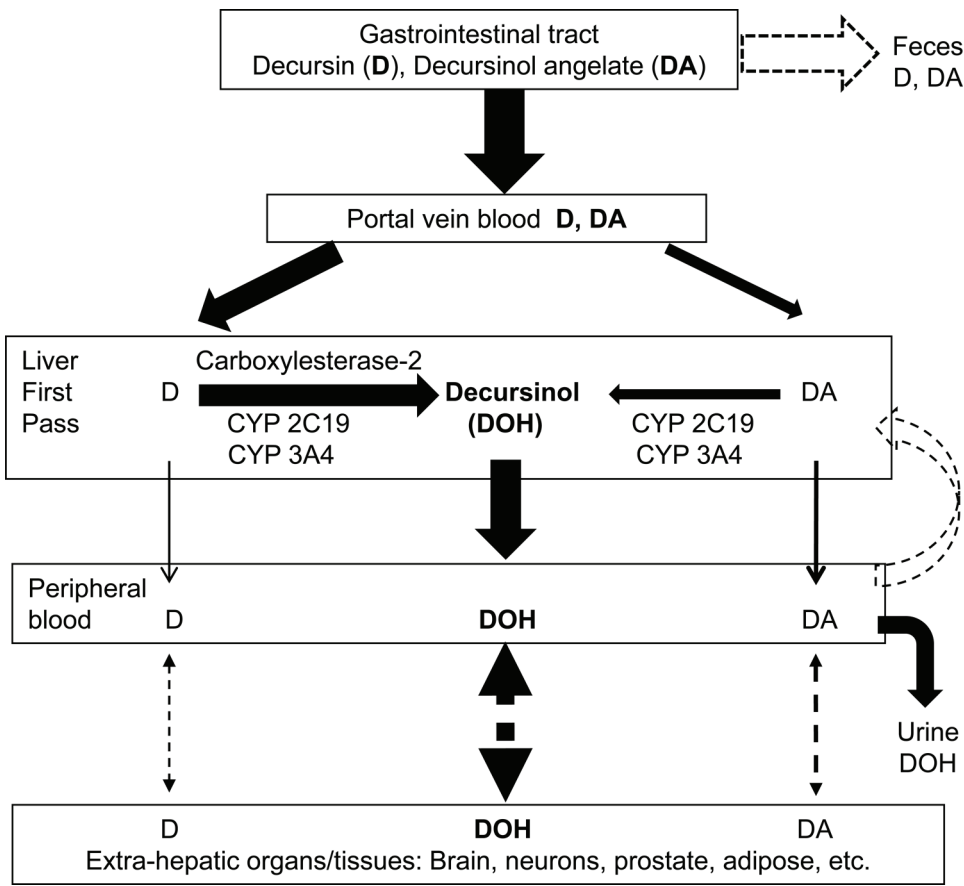


Figure 4. Current knowledge of metabolism of gavage/supplement-delivered pyranocoumarins in rodents and humans. Peripheral peak blood levels of decursin (D) and decursinol angelate (DA) after AGN alcoholic extract oral intake would be expected to be 2–3 orders of magnitude lower (nM to single digit µM) than that of decursinol (DOH) (double to triple digit µM). The portal vein entry of D/DA also applies to *i.p.* injection delivery. CYP—cytochrome P450.

contribution and mechanistic relevance of D and DA and necessitates a better understanding of the contribution and mechanisms of action (MOA) of their common metabolite DOH and the combined actions of expected low level parental “pro-drugs” and higher circulating levels of DOH, with >300 μM achievable, see Zhang *et al.* (2012b) and the following paragraph.

In the earliest reported PK study in an animal model, Park *et al.* (2001) detected DOH ($\sim 80 \mu\text{g}/\text{mL}$, 324 μM) in the plasma of rats after administration of 300 mg/kg body weight of purified D or the methanolic or ether extract of AGN while detecting no D. In our initial PK study of D and DA in mice, we detected DOH as the major metabolite after *i.p.* injection, and the parent compounds were not detected in plasma (Lee *et al.*, 2009). We then systematically evaluated a method for the simultaneous analysis of D/DA and DOH in biological matrixes by HPLC and applied the method to the detection of plasma or tissue levels of these analytes in several *in vivo* studies (Li *et al.*, 2012). Plasma or tumor homogenate was extracted directly with ethyl acetate. In male C57BL/6 mice given a single dose of 4.8 mg D/DA mixture ($\sim 240 \text{ mg}/\text{kg}$) per mouse either by oral gavage or *i.p.* injection, we detected peak plasma concentrations (C_{max}) for D/DA and DOH of 11.2 and 79.7 $\mu\text{g}/\text{mL}$ (46 and 324 μM), respectively, when D/DA mixture was administered *via i.p.* injection; and 0.54 and 14.9 $\mu\text{g}/\text{mL}$ (2 and 60 μM), *via* oral gavage. D/DA and DOH content in the tumor tissues from nude mouse xenografts correlated very well with that in plasma. Overall, our results confirmed that the majority of orally administered D and DA converts to DOH in rodents with a tiny fraction remaining as the parent compounds.

The specific organ site(s) for the conversion of D and DA in rodents is the liver by first pass metabolism (Park *et al.*, 2012a). The plasma concentration of D decreased rapidly after *i.v.* administration in rats with an initial half-life of 0.05 h and was not detectable after 1 h. The concentration of DOH increased rapidly, reaching the peak at $t_{\text{max}} = 0.25 \text{ h}$ and the area under the plasma concentration-time curve (AUC_{DOH}) was 4.4-fold of D (AUC_{D}). In portal vein-cannulated rats, D was the major pyranocoumarin in blood collected from the portal vein after oral D administration, whereas DOH was the dominant form in peripheral blood circulation if D was administered through the portal vein, with the fraction of dose escaping elimination in the liver extremely low ($\text{FH} = 0.11$).

The absorption of DOH itself in rats was quite efficient in a comparison of *i.v.* infusion vs. oral dosing (Song *et al.*, 2011). Following *i.v.* administration, nonlinear elimination of DOH was observed, whereas after oral administration, DOH exhibited rapid absorption (t_{max} 0.4–0.9 h), resulting in a bioavailability of $\sim 45\%$ ($\text{AUC}_{\text{oral}}/\text{AUC}_{\text{iv}}$). We compared the plasma profiles of DOH, when equal molar dosages of D/DA or DOH were given to rats by gavage, and investigated the effect of different excipient formulas and other chemicals in AGN extract on the plasma DOH (Li *et al.*, 2013b). DOH gavage led to a faster attainment of peak plasma DOH ($t_{\text{max}} \sim 0.7 \text{ h}$) and much higher peak levels than an equal molar amount administered as the D/DA mixture or as AGN ethanol extract, resulting in ~ 2.4 -fold higher AUC. Compared to a formula based on ethanol-PEG400-Tween80, carboxyl methyl cellulose was a less efficient vehicle. In addition, we detected peak levels of D and DA with a more sensitive UHPLC-MS/MS method in the plasma of rats administered with D/DA or AGN extract in only the nM range ($t_{\text{max}} \sim 0.5 \text{ h}$). Later on, as we confirmed in a detailed

single-dose PK experiment in mice, DOH dosing in an improved excipient formula led to 3.7-fold plasma AUC_{DOH} over that achieved by equimolar D/DA dosing (Wu *et al.*, 2017).

Contradicting the above, a Korean group reported both D and DA were detectable at high concentrations in rat plasma up to 72 h after the D/DA mixture was administered by either oral gavage (200 mg per rat) or *i.v.* injection (2 mg per rat) (Kim *et al.*, 2009). There were many troubling aspects of this study including the near-total absorption of the orally-delivered dose.

Enzymes for Converting D and DA to DOH and the Role of Cytochrome P450 Isoforms

In test tube assays, Abd El-Aty *et al.* (2008) examined the inhibitory effect of D and DA on hepatic CYP1A1, CYP1A2, CYP2D15, and CYP3A12 catalytic activities using dog liver microsomes. Both D and DA were able to inhibit CYP1A1/2 catalytic activity more than on CYP2D15 and CYP3A12 activities. Enzymatic kinetic studies indicated that D and DA selectively and non-competitively inhibited CYP1A1/2 activity. On the other hand, they exhibited only slight inhibitory effects on CYP2D15 and CYP3A12. Additionally, they showed increased inhibition after pre-incubation, which suggested the involvement of “suicide mechanism”, an irreversible inhibition usually derived from activation of a drug by CYPs into a reactive metabolite, which tightly binds to the enzyme active site, leading to a long lasting inactivation (Fontana *et al.*, 2005). Similarly, suicide mechanism-based inactivation of CYP2A6, the major coumarin 7-hydroxylase in human liver, by DA has been studied by using either human liver microsomes or recombinant CYP2A6 (Yoo *et al.*, 2007). The data indicated that the decrease in the CYP2A6 activity by DA was NADPH-, time-, and concentration-dependent and exhibited a pseudo-first-order kinetic pattern. Pilocarpine, a competitive inhibitor of CYP2A6, dose-dependently attenuated the DA-mediated inactivation of CYP2A6. *In vitro* metabolism was studied by incubating 20 μ M DA with recombinant CYP2A6 in the presence of an NADPH-generating system and revealed that CYP2A6 oxidized DA to the dihydrodiol metabolites, and that the possible epoxide intermediate might account for the inactivation of CYP2A6. Decursin and 7-demethylsuberosin, an important intermediate in the biosynthesis of D and DA (Park *et al.*, 2010), were claimed to have similar inhibitory activity as DA. However, DOH showed no inhibitory effect.

Additional test tube-based CYP isoform activity screening supported D and DA as potential substrates for CYP1A2 and CYP2C19 (Mahat *et al.*, 2013). Another paper reported a potent inhibition of CYP2J2 (which is involved in arachidonic acid metabolism to prostaglandins) by D via a non-competitive mechanism but not by DOH (Lee *et al.*, 2014a). Taking a similar approach, we investigated the metabolic rate of D and DA using liver microsomes from humans and rodents and characterized the major metabolites of D and DA in human liver microsomal incubations using LC-MS/MS (Li *et al.*, 2013a). With liver microsomal incubations of D and DA without NADPH to support CYPs, D was efficiently hydrolyzed to DOH by hepatic esterase(s), which was later identified as carboxylesterase 2 (CES 2), yet DA was not hydrolyzed. In contrast, with the addition of NADPH

to support CYPs, D and DA were both efficiently converted to DOH. Using the LC-MS/MS method, we identified nine intermediate putative metabolites for D and DA, respectively, from human liver microsomal incubations.

To probe for the specific isoforms of carboxylesterases (CES) and CYP enzymes involved, we tested their respective recombinant human isoforms (Zhang *et al.*, 2015b). With CES, D was hydrolyzed by CES2 but not CES1, and DA was resistant to both CES1 and CES2. For CYPs, 2C19 was the most active isoform at metabolizing D and DA *in vitro* followed by 3A4. In human liver microsomal (HLM) preparations, the general CYP inhibitor 1-aminobenzotriazole (ABT) and respective competitive inhibitors for 2C19 and 3A4, (+)-N-3-benzylirivanol (NBN) and ketoconazole, substantially retarded the metabolism of DA and, to a lesser extent, of D. In mice given a single dose of D/DA, pretreatment with ABT boosted the plasma and prostate levels of D and DA by more than an order of magnitude.

Pharmacokinetics of D and DA in Humans

We conducted a single oral dose human PK study of D and DA delivered through an AGN-based dietary supplement CognI.Q (purchased from Quality of Life Labs, Purchase, NY) in 20 healthy subjects, i.e., 10 men and 10 women, each consuming 119 mg D and 77 mg DA from 4 capsules (Zhang *et al.*, 2015a). Analyses of plasma samples using UHPLC-MS/MS showed mean $t_{\max}$ of 2.1, 2.4 and 3.3 h and mean $C_{\max}$ of 5.3, 48.1 and 2,480 nmol/L for D, DA and DOH, respectively. The terminal elimination half-life ($t_{1/2}$) for D and DA was similar (17.4 and 19.3 h), and each was much longer than that of DOH (7.4 h). The mean AUC_{0-48h} for D, DA and DOH was estimated as 37, 335 and 27,579 h·nmol/L, respectively. Gender-wise, men absorbed the parent compounds faster and took shorter time to reach peak DOH concentration. The human data supported an extensive conversion of D and DA to DOH ($AUC_{DOH}/AUC_{total} = 98.7\%$) at the dietary supplement dose tested.

Since we have identified CYP2C19 and 3A4 as the major human isoforms for D and DA metabolism to DOH (Zhang *et al.*, 2015b), we assessed the association of their metabolizer genotypes with key PK metrics. The 2C19 extensive metabolizer genotype (2C19*17 allele) tended to have less plasma DA AUC_{0-48h} and poor metabolizer genotype (2C19*2 allele) tended to have greater DA AUC_{0-48h} . Even with a limited number of subjects, our findings suggest that CYP genotypes and interactions among CYP inhibitor drugs and D/DA should be considered in designing future translational studies of AGN ethanol extracts, DA and/or D.

Adapting our published method, a Korean group established multiple reaction monitoring (MRM) to measure D, DA, DOH, and nodakenin (a furanocoumarin glucoside), respectively (Kim *et al.*, 2018b). They applied it to a PK study for the four coumarins after 10 healthy men were given an oral dose administration of 4.6 g of “AGN” root powder containing 0.055 mg of D, 0.184 mg of DA, and 1.095 mg of nodakenin. They reported values of the elimination half-life ($t_{1/2}$) for D, DA, and DOH similar to each other (3.03, 4.04, and 2.62 h, respectively) but were shorter than $t_{1/2}$ of nodakenin (6.28 h). The peak time $t_{\max}$

values for D and DA (0.44 and 0.31 h, respectively) were similar and slightly sooner than DOH (0.64 h), which was similar to that for nodakenin (0.67 h).

These metrics were much shorter than those reported by us for D, DA, and DOH after oral administration in men ($n = 10$) for AGN-based dietary supplement CognI.Q (119 mg D and 77 mg DA) (Zhang *et al.*, 2015a). The terminal elimination $t_{1/2}$ values for our subjects were D (17.7 h), DA (14.9 h), and DOH (6.6 h). Two prominent differences are noted between these studies. The first was the proportion of D/DA in CognI.Q was roughly 1/0.6 (Zhang *et al.*, 2015a), which has been reported consistently for ethanolic or methanolic extracts of AGN by us and others. The “AGN” powder used to dose the Korean men yielded a reversed D/DA ratio of 1/3.3 (Kim *et al.*, 2018b). The second difference was that the total D + DA dosage for the CognI.Q study was 196 mg, whereas for the Korean trial, it was only 0.24 mg for 4.6 g of AGN powder. The measured D + DA content of 0.0052% in their AGN powder was three orders of magnitude less than that would be typically expected of dried AGN root (3–8.2%). The authenticity of the “AGN” powder utilized for the Korean study is therefore highly questionable.

Knowledge Gaps for PK and Metabolism

There are significant gaps of knowledge in terms of PK and metabolism of the AGN pyranocoumarins. So far, only single dose PK studies have been reported in human subjects (Zhang *et al.*, 2015a; Kim *et al.*, 2018b) with a serious authenticity concern as noted above of “AGN” for the Korean paper (Kim *et al.*, 2018b). The nature of PK dose-response to increasing dosages of AGN supplements is not known and should be studied. In addition, a PK study after repeated AGN exposure in humans as well as in animal models will need to be carried out to determine whether metabolic adaptation occurs over chronic intake, especially whether the extent of D/DA conversion to DOH would be modified. This has significant mechanistic implications for the relative contribution of these molecular entities to a given bioactivity *in vivo*. Furthermore, since hepatic CYPs are principally involved in controlling D/DA conversion to DOH, how CYP geno/phenotypes are related to the above PK metrics will be useful for personalizing dosages to achieve desired health benefits. Moreover, while DOH is established as the major metabolite of D and DA, whether and how it may be further metabolized should be investigated.

Graphic Summary of Reported *In Vivo* Bioactivities of AGN Alcoholic Extracts and Water Extracts

Figure 5A presents a summary of reported (in English literature) *in vivo* bioactivities of AGN alcoholic extracts and their signature compounds against cancers, pain, memory and neurological disorders, ischemic stroke, inflammatory diseases (IBD colitis) and infections (sepsis) and many other diseases and health conditions. Figure 5B summarizes reported *in vivo* bioactivities of AGN water extracts and their sub-fractions. We discuss these *in vivo* bioactivities/evaluations in more detail in the following sections.

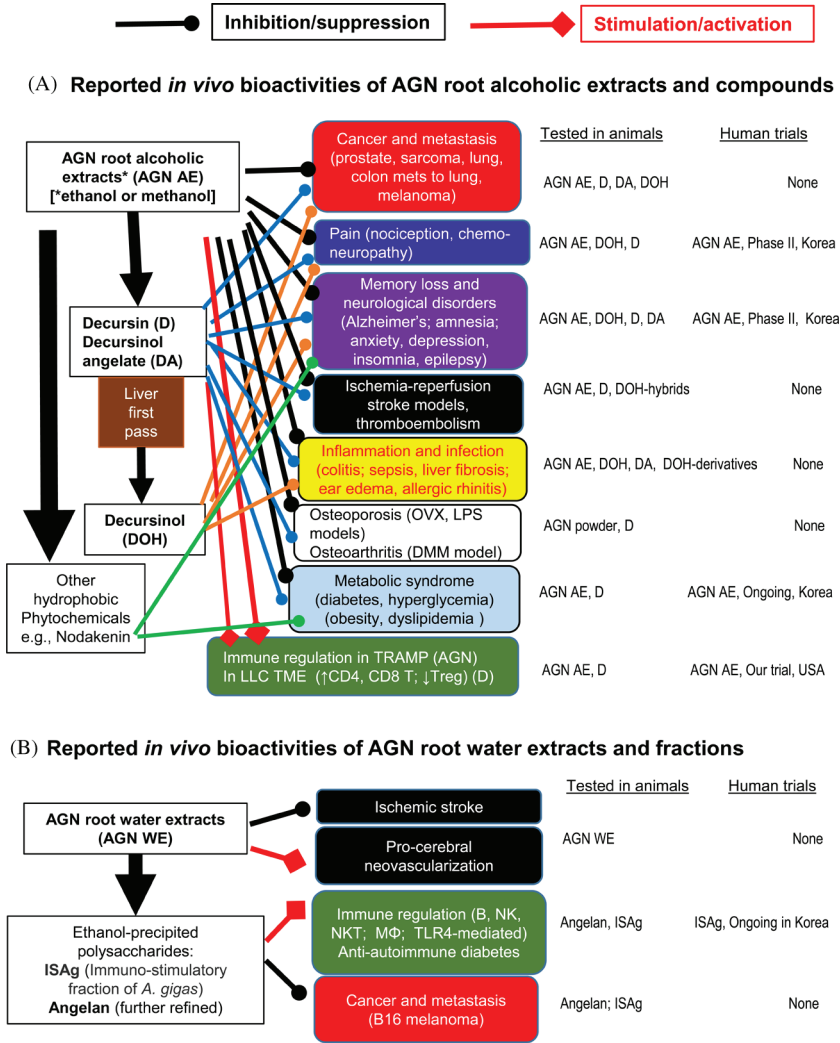


Figure 5. Schematic summary of reported bioactivities of (A) *Angelica gigas* Nakai (AGN alcoholic extracts and signature phytochemicals and (B) water extracts and ethanol-insoluble polysaccharide sub-fractions. Lines with bulb ends show inhibitory/suppression bioactivity. Red lines with diamond ends show stimulatory/activation bioactivity. A solid line indicates an AGN entity has been tested for at least one bioactivity in an animal model.

In Vivo Anticancer Activities of AGN Alcoholic Extracts

Table 1 provides a summary of reported studies in animal cancer models and outcomes. Intervention information included dosage, frequency and route of administration and duration of treatment.

Table 1. Studies of *Angelica gigas* Nakai Root Ethanol Extracts and Signature Pyranocoumarins in Mouse Cancer Models

Mouse Strain	N per Group	Cancer Model	Drug Dose (mg/kg)	Dose Interval	Route	Duration	Anticancer Efficacy	Refs
Mouse, athymic nude	10–15	Xenograft, human DU145 PCA, prophylactic	AGN 100	Once daily	<i>po</i> <i>ip</i>	8 weeks	Tumor wt. –64%; –72%	Lee <i>et al.</i> (2009)
Mouse, athymic nude	10–15	Xenograft, human PC3 PCA, prophylactic	AGN 100	Once daily	<i>ip</i>	8 weeks	Tumor wt. –51%	Lee <i>et al.</i> (2009)
Mouse, SCID NSG	10	Xenograft Human LNCaP-AR PCA, prophylactic	AGN 200, 400 DOH 160	Once daily	<i>po</i>	10 weeks	Tumor wt. –50%; –70% by 2 doses of AGN	Wu <i>et al.</i> (2017)
Mouse, SCID NSG	6–7	Xenograft Human LNCaP PCA, treatment	DOH 120	Once daily	<i>po</i>	4 weeks	Tumor wt. –50%	Lu lab <i>Tbp</i>
Mouse, SCID NSG	17	Xenograft LNCaP PCA, treatment	DOH 120	Once daily	<i>po</i>	4.5 weeks	Tumor wt. –50%	Lu lab <i>Tbp</i>
Mouse, transgenic	15	TRAMP NE carcinoma, prophylactic	AGN 200	Once daily	<i>po</i>	16 weeks	NE Tumor wt. –80%	Zhang <i>et al.</i> (2015c)
Mouse, transgenic	12–20	TRAMP NE carcinoma and metastasis, prophylactic	AGN 200	Once daily	<i>po</i>	8 and 20 weeks	NE Tumor wt. –70% Improved survival to 28 weeks	Tang (2015)
Mouse, transgenic	12–20	TRAMP NE carcinoma and metastasis, prophylactic	D/DA 120	Once daily	<i>po</i>	8 and 20 weeks	NE Tumor wt. –50%. No survival benefit.	Tang <i>et al.</i> (2015)
Mouse, syngeneic	10	Allograft, lung Ca, LLC, prophylactic	D 4	Every other day	<i>ip</i>	2 weeks	Tumor wt. –65%	Jung <i>et al.</i> (2009)
Mouse, syngeneic	6	Allograft, lung Ca, LLC, prophylactic	D or DOH 50	Once daily	<i>ip</i>	2 weeks	Tumor wt. –40%	Lee <i>et al.</i> (2009)
Mouse, syngeneic	5	Allograft, lung Ca, LLC, intervention	D 10	Every other day	<i>ip</i>	9 days	Tumor wt. –40%	Ge <i>et al.</i> (2020)
Mouse, syngeneic	10	Allograft, colon Ca, CT-26 metastasis, prophylactic	D or DOH 10	Once daily	<i>po</i>	2 weeks	Lung metastasis –50%	Son <i>et al.</i> (2011)
Mouse, syngeneic	10	Allograft Sarcoma-180, treatment	D or DA 50, 100	Once daily	<i>ip</i>	9 days, Observe survival	Increased host survival +50%, +77%	Lee <i>et al.</i> (2003)
Mouse, athymic nude	5	Xenograft, B16F10 melanoma intervention	DA 100, 200	Every other day	<i>po</i>	4 weeks	Tumor wt. –40, –64%	Chang <i>et al.</i> (2021)

Notes: *po* – Oral route (gavage, gastric intubation with feeding needle); *ip* – intraperitoneal; D – decursin; DA – decursinol angelate; DOH – decursinol. *Tbp* – to be published from Lu lab.

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Xenograft Models in Immunocompromised Mice

We reported *in vivo* inhibitory efficacy of AGN ethanol extracts on androgen-independent DU145 and PC3 xenografts (Lee *et al.*, 2009) and in LNCaP-derived AR positive models (Wu *et al.*, 2017). For the DU145 xenograft study, mice were treated with 100 mg AGN/kg body weight in 1% Tween 80 by either gavage or *i.p.* injection 7 days before tumor inoculation and continued daily throughout the experiment. By the end of experiment, treatment with AGN by *i.p.* injection and by oral route suppressed the final tumor weight by 73.6% and 63.6%, respectively. None of the treatments negatively affected body weight of the mice. For the PC3 xenograft study, treatment with 100 mg AGN/kg by *i.p.* injection started 4 days after tumor inoculation and continued daily until the end of experiment. The tumor weight in the AGN-treated mice was decreased by 51.2%. Biomarker analyses indicated that antiproliferation, anti-angiogenesis and apoptosis were likely involved in the *in vivo* suppressing effect of AGN against xenograft tumors. In the LNCaP-AR xenograft model, we have shown a dose-dependent suppression of the growth of AR-positive LNCaP-AR *s.c.* xenografts in NSG mice with daily gavage dosage of 5 mg and 10 mg AGN per mouse (~200 and 400 mg/kg body weight) starting 1 week before cancer cell inoculation (Wu *et al.*, 2017). In this model, metastasis to lung and liver was detectable by *ex vivo* bioluminescence of the harvested organs and the metastasis incidence was decreased by AGN in a dose-dependent manner.

Transgenic Mouse Prostate Carcinogenesis Model

We have shown that AGN inhibited the two lineages of carcinogenesis in the prostate of TRAMP (Transgenic Adenocarcinoma Mouse Prostate) mice (Zhang *et al.*, 2015c). In this study, male C57BL/6 TRAMP mice and wild type littermates were given a daily gavage (5 mg/mouse [~200 mg/kg], Monday–Friday) of AGN or 1% Tween 80 vehicle, beginning at 8 weeks of age (WOA). All mice were terminated at 24 WOA, unless earlier euthanasia was necessitated by large tumors. Whereas AGN-treated TRAMP mice decreased dorsolateral prostate (DLP) lesion growth by 30% ($P = 0.009$), they developed fewer and smaller neuroendocrine-carcinomas (NE-Ca) (0.12 g/mouse) than vehicle-treated counterparts (0.81 g/mouse, $P = 0.037$). Proteomic and transcriptomic analyses of banked NE-Ca suggest not only multiple cancer cell targeting actions of AGN but also impacts on the tumor microenvironments (TME) such as angiogenesis, inflammation, and immune surveillance.

We replicated the efficacy outcome for AGN ethanolic extract (a different batch from above; D/DA content as high as 60%) in a follow up experiment in the TRAMP model at a different institution and by different lab personnel (Tang *et al.*, 2015) and compared it side-by-side with D/DA at equimolar dosage. Experimentally, three cohorts of male C57BL/6 TRAMP mice (35–36 mice per group) were fed an AIN93M purified diet and gavaged with a new vehicle (ethanol: PEG400: Tween 80: 5% glucose = 3:6:1:20), AGN (5 mg per mouse) or D/DA (3 mg per mouse, equimolar to that in AGN) 5 days per week from 8 weeks of age (WOA). Mice were euthanized at either 16 WOA ($n = 12$ mice per group) or 28 WOA unless large tumors necessitated earlier sacrifice. Measurement of plasma

and NE-Ca D, DA, and their common metabolite DOH indicated similar retention from AGN vs. D/DA dosing. The growth of TRAMP DLP in AGN- and D/DA-treated mice was inhibited by 66% and 61% at 16 weeks and by 67% and 72% at 28 weeks, respectively. Survival of mice bearing NE-Ca to 28 weeks was improved by AGN, but not by D/DA. Nevertheless, AGN- and D/DA-treated mice had lower NE-Ca burden than the vehicle group. Immunohistochemical and mRNA analyses of DLP showed that AGN and D/DA exerted similar inhibition of TRAMP epithelial lesion progression and key cell-cycle genes. Profiling of NE-Ca mRNA showed a greater scope of modulating angiogenesis, epithelial-mesenchymal transition, invasion-metastasis, and inflammation genes by AGN than D/DA. The data therefore support D/DA as probable active/prodrug compounds against TRAMP epithelial lesions, and they cooperate with non-pyrano-coumarin compounds to fully express AGN efficacy against highly aggressive NE-Ca.

Knowledge Gaps and Challenges

From the cancer model perspective, being extremely aggressive and AR-free and androgen independent, the NE-Ca in the TRAMP mice does not model the predominant adenocarcinomas of human prostate cancer patients which are androgen/AR driven. The xenograft models of human prostate cancer cell lines in athymic nude mice or SCID mice suffered from the incomplete participation of immune cells due to the requisite immune compromised nature of the host mice. Investigations in syngeneic prostate adenocarcinoma/cancer models for the contribution of immune regulation should be planned. There is a paucity of investigation of the AGN extracts in carcinogenesis models of any other organ sites.

Human Clinical Trial Status

No human trial has been reported to assess the utility of AGN alcoholic extract for cancer prevention or treatment. However, there were case documentations of AGN use along with 9 other herbs in 2 cancer patients in Korea (Lee *et al.*, 2020a) with a remarkable prostate cancer antigen (PSA) suppression response in the male patient and improvement of quality of life for both.

Anticancer Activities of D, DA and DOH

Cell culture models have been primarily used as screening tools as well as cellular targets for studying potential anticancer activities and “mechanisms” of these compounds. Indeed, we discovered the potent anti-AR signaling activity of D and DA at exposure concentrations that were at least 1 order of magnitude lower than apoptotic level using the LNCaP cells as the *in vitro* screening model (Jiang *et al.*, 2006; Guo *et al.*, 2007) and examined the likely structure activity relationship (SAR) of D, DA and DOH. DOH exhibited a bi-phasic effect on AR signaling in the presence of androgen and a partial agonist role in the absence of androgen (Guo *et al.*, 2007). At the single micromolar subapoptotic concentrations,

D and DA promoted neuroendocrine differentiation of LNCaP cells to distinctive neurite bearing morphology. This SAR applied to estrogen-dependent breast cancer MCF-7 cells as well as estrogen receptor negative MDA-MB-231 breast cancer cells (Jiang *et al.*, 2007). Cell culture models of other organ sites included lung, colon, bladder, sarcoma and blood cancers and were extensively reviewed in our paper a decade ago (Zhang *et al.*, 2012a). Additional cancer cell lines have been studied and reported since, yet as a pessimistic commentary, without a justification of relevance of pyranocoumarin exposure levels and with a seeming ignorance of the extensive PK and metabolism knowledge as presented in the previous section (as summarized in Fig. 4). Here, we will focus on discussing those organ sites with available *in vivo* efficacy information: prostate cancer, lung cancer, colon cancer, sarcoma and melanoma.

Prostate Cancer

As mentioned above, we compared D/DA mixture at equimolar dosage to AGN dose of 5 mg per mouse in the TRAMP model (Tang *et al.*, 2015). Measurement of plasma and NE-Ca D, DA, and their common metabolite DOH indicated similar retention from AGN vs. D/DA dosing. The growth of TRAMP dorsolateral prostate (DLP) was equally and substantially inhibited by AGN or D/DA. Survival of mice bearing NE-Ca to 28 weeks was improved by AGN, but not by D/DA. Profiling of NE-Ca mRNA showed a greater scope of modulating angiogenesis, epithelial–mesenchymal transition, invasion–metastasis, and inflammation genes by AGN than D/DA. The data therefore support superiority of AGN to D/DA against NE-Ca, and that D/DA were probable “active” prodrug compounds through DOH against TRAMP epithelial lesions.

Taking advantage of the cryopreserved DLP and NE-Ca tissues, we compared by microarray hybridization the mRNA levels in pooled DLP tissues and individual NE-Ca to characterize potential molecular targets of AGN extract and D/DA (Tang *et al.*, 2021). Clustering and principal component analyses supported distinct gene expression profiles of TRAMP DLP vs. NE-Ca. Pathway Enrichment, Gene Ontology and Ingenuity Pathway Analyses of differential genes indicated that AGN and D/DA affected chiefly processes of lipid and mitochondrial energy metabolism and oxidation–reduction in TRAMP DLP, while AGN affected neuronal signaling, immune systems and cell cycling in NE-Ca. Protein–Protein Interaction Network analysis predicted, and reverse transcription-PCR verified, multiple hub genes common in the DLP of AGN- and D/DA-treated TRAMP mice at 28 WOA and selected hub genes attributable to the non-D/DA AGN components. The vast majority of hub genes in the AGN-treated NE-Ca differed from those in TRAMP DLP. The transcriptomic approach illuminated vastly different signaling pathways and networks, cellular processes and hub genes for the two TRAMP prostate malignancy lineages, and their associations with the interception efficacy of AGN and D/DA.

We further tested whether DOH might be an *in vivo* active compound for suppressing xenograft growth of human prostate cancer cells expressing AR (Wu *et al.*, 2017). In SCID-NSG mice carrying subcutaneously inoculated human LNCaP/AR-Luc cells over-expressing the wild type AR, we compared the efficacy of 4.5 mg DOH per mouse with an

equimolar dose of 6 mg D/DA per mouse. The result showed that DOH decreased xenograft tumor growth by 75% and the associated lung metastasis, whereas D/DA exerted little effect. Measurement of plasma DOH concentration, at 3 h after the last dose, showed a higher circulating level in the DOH-treated NSG mice than in the D/DA-treated mice. A subsequent single-dose PK experiment showed that DOH dosing led to 3.7-fold area under curve (AUC) of plasma DOH over that achieved by equimolar D/DA dosing.

In yet to be published work of Kim *et al.* we evaluated the interception efficacy of DOH against established LNCaP xenografts in NSG mice and assessed the involvement of the AR-PSA signaling axis. In two animal experiments, daily gavage of DOH (120 mg/kg body weight) for 1 month in NSG mice bearing established subcutaneous xenograft tumors suppressed tumor growth by half, without decreasing body and prostate weights. Immunohistochemistry staining for AR and prostate specific antigen did not detect a reduction of either protein in the responder tumors. Nor was there a reduction of their mRNA levels in both the responder and non-responder tumors per RNA seq. These results support DOH as an active compound to exert *in vivo* prostate cancer growth and metastasis inhibitory activity in the preclinical models, independent of AR-axis signaling, which served as the primary screening “target” in our original cell culture studies (Jiang *et al.*, 2006; Guo *et al.*, 2007). Available data support that other phytochemicals in AGN ethanol extracts contribute to or synergize with the pyranocoumarins for the broader scope of anti-prostate cancer action and host survival advantage than the isolated compounds.

Sarcoma

In a mouse allograft study, D or DA, at doses of 50 and 100 mg/kg by *i.p.* injection daily for 9 days, significantly suppressed Sarcoma-180 tumor growth in syngeneic ICR mice (Lee *et al.*, 2003). Compared with the median survival time for the control group of 22 days, those for 50 and 100 D mg/kg treated groups were 29 and 40 days, and those for 50 and 100 mg DA/kg treated groups were 32 and 34 days, respectively.

Lung Cancer

In the experimental LLC lung cancer allograft model, D or DA exerted notable growth inhibitory activity. Jung *et al.* (2009) reported anti-neo-angiogenesis activity of D and DA and attributed this effect to inhibition of VEGFR2 signaling by these compounds. The microvessel density in tumors treated with D (*i.p.*, 4 mg per kg, every other day) for 14 days was significantly decreased compared with a vehicle control group. In cell culture, D and DA inhibited VEGF-induced phosphorylation of VEGFR-2, extracellular signal-regulated kinases and c-Jun N-terminal kinase mitogen-activated protein kinases.

Ge *et al.* (2020) recently reported immune involvement in the hypoxic tumor microenvironment (TME) by D in the LLC allograft model. Treatment with D (*i.p.*, 10 mg/kg, every other day) reduced tumor size and the hypoxic area and HIF-1 α and PD-L1 expression. IHC detected increased infiltrating T cells (CD3+), helper T cells (CD4+) and cytotoxic (CD8+) T cells and fewer regulatory T cells (Foxp3) and myeloid-derived suppressor cell-mediated

immune suppressors (arginase-1 positive). In cell culture with a human lung cancer cell line, they showed that D inhibited HIF-1 activation under hypoxia by down-regulating the protein level of its subunit HIF-1 α and increased oxygen-dependent hydroxylation and ubiquitination of HIF-1 α to promote HIF-1 α degradation and decreased mRNA expression of HIF-1 α target genes. A direct comparison with DOH *in vivo* would have provided more insight into whether these cited actions were relevant to *in vivo* efficacy.

Murine Colon Cancer Metastasis to Lung

In a CT-26 murine colon cancer model, oral administration of D and DOH (10 mg/kg) given 30 min before the tail vein injection of the cancer cells and followed by daily oral dosing of D or DOH for 14 days reduced the number of tumor nodules in the lungs and blunted the increase in lung weight caused by CT-26 metastases (Son *et al.*, 2011). The *in vitro* modeling of the effects of D and DOH in cell culture revealed decreased expression of matrix metalloproteinase (MMP)-2 and MMP-9 in cells and their activities in the culture medium with minimal effect on cell viability in the exposure range up to 20 μ M. Studies with inhibitors for the extracellular signal-regulated kinase (ERK), c-Jun N-terminal kinase (JNK), and phosphatidylinositol-3 kinase (PI3K) supported ERK and JNK-driven MMP-9 playing a critical role for the invasion of CT26 cells. They observed that D and DOH down-regulated ERK and JNK phosphorylation. As the PK studies have supported a pro-drug role of D and DA for DOH, DOH may be the principal *in vivo* chemical mediator for targeting MMPs and their upstream signaling molecules.

Melanoma

A recent paper (Chang *et al.*, 2021) tested DA by gavage administration for efficacy against murine melanoma cell line (B16.F10) *s.c.* xenograft in athymic nude mice. Starting treatment after tumors had reached 50 cubic mm 3 times per week for 4 weeks inhibited tumor weight in a dose-dependent manner at 100 and 200 mg/kg dosages. Biomarker analysis of the tumors showed dose-dependent increase of apoptosis related to mitochondria intrinsic signaling (cytochrome C, Bax/Bcl2 proteins). Extensive cell culture modeling with up to 100 μ M DA indicated anti-autophagic signaling (decreased ATG5, ATG7, beclin-1, LC-3 lipid modification) along with ROS-mediated apoptosis. However, the relevance of such mechanistic findings in the cell culture modeling with unreasonably high exposure concentrations of the pro-drug DA should always be suspected.

Knowledge Gaps and Challenges

As the PK and metabolism studies of D and DA have supported their pro-drug role and rapid conversion to DOH, DOH is the probable principal *in vivo* chemical mediator for their anticancer activities. With a paradoxical lack of direct cytotoxic and cytostatic activity of DOH against cancer cells in cell culture (Yim *et al.*, 2005; Jiang *et al.*, 2006; Guo *et al.*, 2007; Jiang *et al.*, 2007), future cell culture modeling should emphasize the *in vivo*

achievable concentrations of each entity in order to be relevant, i.e., high and varying DOH vs. low and limited D/DA concentration range and with non-cancer cells that make up the tumor microenvironments (TME), including inflammatory and immune cells. Depending on the dosage and metabolic adaptation upon chronic intake, if any, the overall anticancer efficacy outcome would be expected to be the integrated actions of the three pyranocoumarin species.

Human Clinical Trial Status

No human trial has been reported or registered to assess the utility of pyranocoumarins for cancer prevention or treatment.

Anticancer Activities of Other AGN Phytochemicals

We reviewed phytochemicals other than D and DA isolated from AGN that have been evaluated by the NCI Developmental Therapeutics Program for the *in vivo* ability to increase survival of allograft tumor-bearing syngeneic mice (http://dtp.nci.nih.gov/docs/dtp_search.html) (Zhang *et al.*, 2012a). The furanocoumarin xanthotoxin (also known as 8-methoxypsoralen and methoxalen, see structure in Fig. 1C) exerted a potent anticancer activity in 25–64 mg/kg dosage range that achieved increased survival time of the treated mice. By comparison, its lethal dose was 256 mg/kg. Known for photo-sensitization and anticancer activities (Pathak and Fitzpatrick, 1992), xanthotoxin is present in AGN root in trace amount. Its contribution to the anticancer activity of AGN extract is therefore uncertain. Scopoletin was much less potent (400 mg/kg) without detectable lethality in the tested dose range to the host mice. Marmesin, umbelliferone and bergapten treatment did not result in a survival advantage in the dosage ranges tested.

Marmesinin, nodakenin, skimmin, apiosylskimmin and magnolioside showed no direct cytotoxicity ($IC_{50} > 100 \mu M$) toward the SK-OV-3 ovarian cancer cell line *in vitro* (Moon *et al.*, 2011). Marmesinin and magnolioside showed antiplasmodial activity against chloroquine-sensitive strains of malaria pathogen *Plasmodium falciparum* with IC_{50} of 5.3 and 8.2 μM , respectively (Moon *et al.*, 2011). There are no reported evaluations of *in vivo* anticancer activity for these compounds in animal models.

Immune Regulatory and Anticancer Activities of AGN Water Extracts

Compared to alcoholic extracts, much less has been published with AGN water extracts or their sub-fractions, as outlined in Fig. 2 schema. Their reported bioactivities are summarized in Fig. 5B.

Angelan

Angelan is named for a refined sub-fraction of a water extract of AGN root that was precipitated out with ethanol, containing mostly polysaccharides (90%) (Han *et al.*, 1998).

Angelan is approximately 10-kDa in size and mainly composed of arabinose, galactose and galacturonic acid but not glucose (Han *et al.*, 1998). It has been reported to lengthen the survival time of B16F10 melanoma-bearing syngeneic mice when administered *i.p.* at a dose of 30 mg/kg (Han *et al.*, 2006). The primary targets of angelan were assumed to be macrophages and natural killer (NK) cells, which were involved in non-specific innate immunity against cancer cells or virally-infected cells (Han *et al.*, 1998; 2006). In a cell culture model, angelan markedly induced transcription of inducible nitric oxide synthase (iNOS), interleukin-1 β (IL-1 β) and tumor necrosis factor α (TNF- α) in murine macrophage RAW 264.7 cells, mediated by specific activation of NF- κ B/Rel (Jeon *et al.*, 1999; 2001). Another study claimed the macrophage activation by angelan differed from that induced by bacterial lipopolysaccharide (LPS) in three aspects (Jeon *et al.*, 2000). Further studies indicated that angelan activated NF- κ B/Rel through the CD14 and complement receptor type 3 (CR3) membrane receptor and p38 mitogen activated protein kinase (P38MAPK), the latter is critical for the signal transduction leading to NF- κ B/Rel activation in murine macrophages (Jeon and Kim, 2001). Subsequently, the same group investigated the effects of angelan on dendritic cell (DC) maturation and found a crucial role of toll-like receptor 4 (TLR4) signaling pathways (Kim *et al.*, 2007b). These results suggested the possible use of angelan for activating DC-based immunotherapies.

ISAg

Although prepared with a similar procedure, ISAg (immuno-stimulatory component of *A. gigas*) was a less refined extract preparation than angelan and differed from angelan in the makeup of the polysaccharide composition (Kim *et al.*, 2018a), such that glucose constituted 70% of total carbohydrate in ISAg. ISAg also contained more protein than did angelan (17% vs. 8%). In the murine macrophage RAW264.7 cell culture model, ISAg treatment induced nitric oxide (NO) production and cytokine gene expression involved in innate immune responses much like angelan and LPS.

Oral gavage in the dose range of 1, 2 and 4 mg per mouse, every other day for 4 weeks activated macrophages and DCs to secrete cytokine IL-12 (Kim *et al.*, 2018a). Because IL-12 plays a crucial role in NK and NKT cell activation, the authors reported increased activation of NK and NKT cells producing IFN- γ and TNF- α by ISAg oral supplement in the mice. They established that these immune stimulatory effects were mediated through the TLR4 and IL-12 signaling pathway as supported by the attenuated responses to ISAg in the TLR4 mutant- or IL-12p35-knockout mice. In syngeneic mice bearing *s.c.* B16 melanoma cell allografts, the oral gavage treatment with 4 mg ISAg 3 times per week for 2 weeks decreased tumor weight by more than 60%. The efficacy was accompanied by increased infiltrating NK and NKT cells in the tumors.

In a follow up study, the same group asked whether the use of ISAg *in vivo* can affect the differentiation of conventional T cells (Lee *et al.*, 2019a). With respect to CD4⁺ and CD8⁺ T cells, they found that the oral administration of 4 mg ISAg 3 times per week for 4 weeks induced the polarization of CD4⁺ T cells toward the acquisition of the Th1 phenotype *in vivo*. Additionally, in mice treated with ISAg, CD8⁺ cytotoxic T cells produced more IFN γ than in control mice treated with PBS. Moreover, treatment with ISAg activated

CD4⁺ and CD8⁺ T cells as well as NK and NKT cells, resulting in the secretion of Th1-type cytokines in a TLR4-dependent manner. ISAg treatment increased the number of Foxp3⁺ Treg cells, but not Th2 cells, compared to control mice treated with PBS, indicating that ISAg possesses an immunomodulatory capacity that could also control adaptive immune responses in addition to the innate NK and NKT cells. Taken together, the results indicated that ISAg possessed a Th1-enhancing activity that could be used to treat Th2-mediated allergic immune diseases such as atopic dermatitis. However, how much of the immune stimulating substance like angelan or ISAg is present in the alcoholic extract of AGN sold as dietary supplements is not known.

Human Clinical Trial Status

No human trial has been reported or registered to assess the utility of AGN water extract for cancer prevention or treatment or immune regulatory effects. We have started a small-scale clinical trial in healthy men to test the immune regulatory effect of AGN ethanol extract (NCT03630328, PI: Lu). A randomized, placebo-controlled, double blinded, cross over design is used to test the safety and the impact of AGN (CognI.Q) dietary supplement at the vendor recommended dose (i.e., two 200-mg capsules, twice daily, 800 mg total) on peripheral NK and other immune phenotypic and functional metrics.

Analgesic (Pain Killing) Effects

AGN extract supplements are sold for minor aches and pain in the US as *Decursinol-50* and several other brand names. Although human clinical trial and use experience have been cited in product websites (e.g., https://www.lifeextension.com/magazine/2007/7/report_korean_angelica), the results were not published in a peer-reviewed journal. Animal models have been studied for a range of pain afflictions and possible mechanisms.

AGN Extract Against Nociceptive Pain Models

Choi *et al.* (2003a) first reported in the English literature an antinociceptive (pain-killing) profile of AGN ethanol extract in various mouse pain models such as tail-flick and hot-plate tests, acetic acid-induced writhing and intra-plantar (hind paw) formalin tests with various doses (from 0.25 to 3 g/kg body weight by oral dosing). Additional pain models included pro-inflammatory cytokine-induced, substance P- and capsaicin-induced nociceptive tests, as well as excitatory amino acids-induced nociceptive test. They used the Rotarod test to rule out the possibility that antinociceptive effect was caused by motor dysfunction or paralysis. They showed that AGN delayed latencies of the tail-flick and hot-plate paw-licking responses in a dose-dependent manner. In *i.p.* acetic acid-induced writhing test, AGN dose-dependently decreased writhing numbers. Furthermore, the cumulative response time of nociceptive behaviors induced by intraplantar formalin injection was reduced during both the first (acute) and the second (inflammation) phases

in dose-dependent manners. Moreover, oral administration of AGN blunted licking, scratching and biting responses induced by TNF- α (100 pg), IFN- γ (100 pg) or IL-1 β (100 pg) injected *intrathecally* (*i.t.*), especially at the highest AGN dose (3 g/kg), as well as by substance P or capsaicin. Finally, oral administration of AGN decreased the nociceptive responses elicited by *i.t.* injection of glutamate (20 μ g), N-methyl-D-aspartic acid (60 ng), α -amino-3-hydroxy-5-methylisoxazole-4-propionic acid (13 ng) or kainic acid (12 ng). These results suggested that AGN exerted antinociception via acting on the central nervous system and showed antinociceptive profiles in various pain models, especially inflammatory pain.

Pain Killing Effect of DOH and Mechanistic Insights

The same group reported the antinociceptive mechanisms of orally-administered DOH (5 to 200 mg/kg body weight, in an ethanol/polyethylene glycol 400/sodium carboxymethyl-cellulose-based vehicle) in ICR mice (Choi *et al.*, 2003b). The results indicated that DOH showed dose-dependent antinociceptive efficacy in the various pain models (ED₅₀ ~50 mg/kg, peak time 30 min post dose). Moreover, the cumulative response time of nociceptive behaviors induced by intraplantar injection of formalin and intrathecal injection of TNF- α , IL-1 β , IFN- γ , substance P or glutamate, as in the study above, was dose-dependently diminished by DOH. Further mechanistic studies using receptor antagonists suggested the antinociception of DOH could be mediated by noradrenergic, serotonergic, adenosine A₂, histamine H₁ and H₂ receptors. Interestingly, the pain killing effect was not modified by an opioid receptor antagonist, naloxone and, therefore, was not likely mediated through opioid receptor signaling pathways.

Later on, Seo *et al.* (2009) studied the analgesic effect of DOH using the acetic acid-induced writhing test in male ICR mice. In their study, the antinociceptive effects of DOH were observed at an orally-administered dose of 50 mg/kg but not at 25 or 10 mg/kg, more potent than analgesic effects of aspirin (acetyl salicylic acid) and acetaminophen (Tylenol), each efficacious at an orally-administered dose of 200 mg/kg but not at 50 or 100 mg/kg. A synergistic effect between DOH and Tylenol was observed in the group co-administered DOH (25 mg/kg) and Tylenol (100 mg/kg). These results were interpreted to indicate that the analgesic effect of DOH involved supraspinal cyclooxygenase regulation that might overlap with acetaminophen-induced analgesic mechanisms rather than systemic or peripheral prostaglandin modulation by aspirin.

Chemotherapy-Induced Neuropathic Pain (CINP) Relief by D (Modeling Epidural Injection)

In contrast to the conventional herbal oral supplement approach for pain relief/testing, a 2021 paper reported alleviation of CINP in a mouse model by intrathecal injection of D (Son *et al.*, 2021). CINP is a severe adverse effect of platinum- and taxane-derived anticancer drugs and is often dose-limiting in patients. The authors established a mouse model of paclitaxel-induced peripheral neuropathy and examined the impact of intrathecal injection

of D (50 mg/kg) on the resulting CINP. Paclitaxel (2 mg/kg, diluted with phosphate-buffered saline) was *i.p.* injected four times at three-day intervals. The mechanical allodynia (increased pain sensation) test was carried out by the von Frey method on days 7 and 14 since the first injection of paclitaxel. Two groups ($n = 12$ each) of chosen CINP mice were each given an intrathecal injection (L5–L6 intervertebral space) of 5 μ L of D or vehicle three times every other day, for 6 days, to examine whether D could mitigate mechanical allodynia. There was no relief of mechanical allodynia by the first injection of D, yet 30 min after the second injection and third injection 2 days apart, the allodynia was effectively blunted, and the relief of pain lasted up to 3 h. We speculate that the lack of any allodynia relief by the first dose could be related to a lack of D conversion to DOH in the spinal column, and the first exposure could have induced CYP or CES to allow subsequent doses of D to efficiently produce DOH to exert the pain relief observed after the second and third doses. This hypothesis is yet to be tested.

Since the pathophysiology of CINP involves damage to neuronal networks and dysregulation of signal transduction due to abnormal Ca^{2+} levels, the authors used a cell culture model of F11 cells (a somatic cell hybrid of a rat embryonic dorsal root ganglion [DRG] and mouse neuroblastoma cell line N18TG) to test the direct impact of D on capsaicin-induced rise of intracellular Ca^{2+} level and neurite outgrowth. They observed a concentration-dependent response in each metric within the D exposure concentration up to 2 μ M. Subsequently, they discovered that D interacted with the transient receptor potential vanilloid 1 (TRPV1) as a potent antagonist with $\text{IC}_{50} \sim 1 \mu\text{M}$ and such level of D exposure activated AKT and CREB phosphorylation. Therefore, D might be an important compound for the treatment of paclitaxel-induced neuropathic pain via TRPV1 inhibition and recovery of damaged neuronal networks.

Although not directly translatable to the setting of an herbal supplement modality, the study has significant mechanistic and clinical pain management implications. Through intrathecal delivery to bypass the liver CYPs and CES2 (Fig. 4), the model assessed the direct impact of D on neuronal signaling related to CINP *in vivo*. The cell culture study was well conceived to model with a non-cytotoxic level of D ($\leq 2 \mu\text{M}$) that could be achievable through an oral supplement route in spite of liver first-pass catabolism. It is desirable to replicate the study of DOH on these *in vivo* and *in vitro* metrics to dissect the targeting specificity and contributions of this hepatic metabolite and the parental drugs D and DA.

Knowledge Gaps

So far in different nociceptive pain models, the efficacy of AGN or DOH appeared to be full range, while the potency appeared to be a magnitude less than the classic opiate morphine in the same tests ($\text{ED}_{50} \sim 3 \text{ mg morphine/kg}$ by *i.p.* route). As its analgesic effect was not likely mediated through opioid μ -receptor (Choi *et al.*, 2003b), the potential for using DOH or AGN supplement to spare opioid drugs should be evaluated. As DOH is a known inhibitor of acetylcholine esterase (AChE) (Kang *et al.*, 2001), the observation that another AChE drug donepezil attenuated morphine tolerance in rat (Sharifipour *et al.*, 2014) prompts a speculation of DOH for interfering with opioid tolerance and should be

tested. No studies have been reported concerning tolerance profiling of DOH in pain models and should be studied.

Human Clinical Trial Status

Although a human clinical trial in Korean patients has been cited in product websites (e.g., https://www.lifeextension.com/magazine/2007/7/report_korean_angelica), the results were not published in a peer-reviewed journal. Randomized, blinded, placebo-controlled trials are warranted to test the safety and efficacy in a range of pain conditions, especially incorporating AGN supplement into standard of care pain management regimen for opioid sparing and improvement of quality of life of the patients.

Cognitive/Memory Health

AGN ethanol extract trade-named INM-176 is the active herbal ingredient for supplement product CognI.Q marketed in the US (<https://qualityoflife.net/>). A phase II clinical trial in elderly Korean patients with memory impairment (Alzheimer's dementia) had been published in Korean (Kim *et al.*, 2003) with promising efficacy potential. However, the results of a follow-up trial comparing AGN to donepezil, an AChE inhibitor drug, have not been reported (Clinicaltrials.gov NCT01245530). Considerable animal modeling work around the turn of the millennium provided the rationale for that first human clinical trial. Additional published animal studies provided further support for potential cognitive/memory benefit of AGN alcoholic extracts and the pyranocoumarins.

DOH Identified as an AChE Inhibitor by Activity-Guided Fractionation

As the neurotransmitter acetylcholine was known to decline in Alzheimer's patients, Kang *et al.* (2001) identified DOH from an AGN methanolic extract as the most potent AChE inhibitor in an activity guided fractionation study that tested 12 coumarins. The IC_{50} for pyranocoumarins DOH, xanthyletin (from dehydration/oxidization of DOH 3'OH), and D was 28, 150, 390 μ M, respectively, and that for the furanocoumarins marmesin, xanthotoxin, isoimperatorin and nodakenin was 67, 54, 69 and 68 μ M, respectively. As a reference, the AChE drug donepezil and the phytochemicals berberine and galantamine were more potent with reported IC_{50} of 0.057, 3.1 and 4.2 μ M, respectively (Amat-Ur-Rasool *et al.*, 2021).

AChE Inhibitory Potency of DOH and D Did Not Correlate with Anti-Amnestic Activity in Scopolamine Model

The same group (Kang *et al.*, 2003) evaluated the anti-amnestic activity of D *in vivo* using ICR mice with amnesia induced by scopolamine (1 mg/kg body weight, s.c.), a known anticholinergic drug for treating motion sickness and post-operative nausea and vomiting.

D, when administered at 1 and 5 mg/kg body weight *i.p.* 60 min before scopolamine, significantly ameliorated scopolamine-induced amnesia as measured in the passive avoidance test, but DOH in the same dose range was only ¼ as efficacious. D at 1 mg/kg dose antagonized scopolamine-induced loss of both working (short-term) and reference (long-term) memory in the Morris water maze test. Moreover, D inhibited AChE activity by 34% in the hippocampus of treated mice, but not other brain regions tested.

β-Amyloid Memory Loss Prevention

Yan *et al.* (2004) reported a protective effect of long-term oral administration of the ethanol (95%) extract of AGN or purified DOH on β-amyloid peptide 1–42-induced memory impairment in mice. The efficacious dose and pretreatment time required for AGN extract were first determined through drinking water containing different concentrations of AGN extract (0, 0.08 vs. 0.1%, w/v). After 1, 2, and 4 weeks, Aβ(1–42) (410 pmol) was administered via intracerebroventricular injection. Pretreatment of mice with AGN extract (0.1% in drinking water) for 4 weeks significantly blocked the Aβ(1–42)-induced impairment in passive avoidance performance, and pretreatment for shorter period (1 or 2 weeks) were not protective statistically. When mice were fed chow diet mixed with various doses of DOH (dissolved in corn oil and mixed into diet, 0.001%, 0.002%, and 0.004%, w/w) for 4 weeks before intracerebroventricular injection of Aβ(1–42), all regimens significantly attenuated the Aβ(1–42)-induced impairment in passive avoidance performance. Only the highest dose of DOH (0.004%) significantly blunted the Aβ(1–42)-induced decrease in alternation behavior (spatial working memory) in the Y-maze test without change in general locomotor activity. These findings supported DOH as an, if not the, active compound of AGN for preventing memory impairment related with Aβ Alzheimer's disease.

Comparison of AGN vs. AChE Inhibitor Drug Donepezil

As mentioned above, INM-176 is a standardized ethanolic extract of AGN root and is used exclusively in herbal supplement products sold in the USA. Its efficacy and mechanism of action was compared with donepezil in the following two papers from the same research group. Park *et al.* (2012c) confirmed that INM-176 exhibited anti-amnesic effects in the scopolamine- or Aβ(1–42)-induced memory impairment using the passive avoidance and the Morris water maze tasks in mice. Scopolamine-induced cognitive dysfunction was significantly attenuated by a single or sub-chronic administration of INM-176 in the passive avoidance and the Morris water maze tasks. A single or sub-chronic administration of INM-176 also ameliorated memory impairments induced by Aβ(1–42) protein. While not as potent as donepezil (5 mg/kg donepezil vs. ~300–600 mg/kg AGN), AGN was as efficacious or superior for the maximal extent of benefit. INM-176 inhibited AChE activity in the hippocampal tissue *in vitro* and *ex vivo*. In addition, INM-176 attenuated the Aβ(1–42) protein-induced astrocyte activation in the hippocampus as well as cholinergic neuronal

damage in the CA3 region of the hippocampus and the nucleus basalis of Meynert. These results affirmed the ameliorating effects of INM-176 on scopolamine- or A β (1–42) protein-induced memory impairment in association with AChE inhibition and neuroprotective activities.

Neuroinflammation Memory Loss Protection

Park *et al.* (2012b) further investigated whether INM-176 exhibits neuroprotective activities against lipopolysaccharide (LPS)-induced neuronal damage. LPS (1 μ g/2 μ l, *i.c.v.*)-induced cognitive dysfunction in mice, as determined by passive avoidance and Y-maze tasks, was significantly attenuated by the administration of INM-176. Furthermore, the activation of microglia or astrocytes by LPS in the hippocampal regions of mice was suppressed by INM-176. These results suggest that the neuroprotective and cognition ameliorating effects of INM-176 against LPS-induced damage are mediated, in part, by its anti-inflammatory activities.

Mild Brain Injury and Chronic Stress Memory Loss Prevention

Kim *et al.* (2019) evaluated the memory enhancing effects of an AGN ethanolic extract in mouse models of mild cognitive impairments based on mild traumatic brain injury and chronic mild stress. AGN ethanolic extract given at 1 mg/kg dosage orally improved spatial learning, working memory, and suppressed fear memory in the mild traumatic brain injury model. It also improved spatial learning and suppressed cued fear memory in the chronic mild stress model animals. However, it should be cautiously noted that the AGN dosage reported appeared to be 2 or 3 orders of magnitude lower than those reported by other groups for many studied bioactivities.

Novel Whole Root Powder Formulation on Memory Health

Taking a different formulation approach with AGN whole root fine powder, another group replicated the oral absorption and memory enhancing effect (Piao *et al.*, 2015). Oral solid formulations based on AGN and Soluplus were prepared by the hot-melting extrusion (HME) method. Ultrafine AGN particles were used in the HME process with high shear to produce AGN-based formulations. In simulated gastrointestinal fluids (pH 1.2 and pH 6.8) and water, significantly higher amounts of D and DA were extracted from the HME-processed AGN/Soluplus (designated as F8) group than from the AGN ethanol extract ($p < 0.05$). Based on the PK study, the relative oral bioavailability (AUC_{DOH}) in F8-administered mice was 8.75-fold higher than in AGN extract-treated group. In scopolamine-induced memory-impaired mice, F8 exhibited a more potent cognitive enhancing effect than AGN extract in both the Morris water maze test and passive avoidance test. These findings suggest that HME-processed AGN/Soluplus formulation (F8) could be an alternatively manufactured efficacious therapeutic candidate for memory impairment.

Memory Promotion by Non-Pyranocoumarins in AGN

In addition to pyranocoumarins, nodakenin, a furanocoumarin glucoside compound (Fig. 1C), also ameliorated scopolamine (1 mg/kg, *i.p.*)-induced memory disruption in mice (Kim *et al.*, 2007a) in the passive avoidance test, the Y-maze test, and the Morris water maze test. Nodakenin (10 mg/kg, *p.o.*) administration significantly reversed scopolamine-induced cognitive impairments in the passive avoidance test and the Y-maze test ($P < 0.05$) and also reduced escape latency during training in the Morris water maze test ($P < 0.05$). Moreover, swimming times and distances within the target zone of the Morris water maze were greater in the nodakenin-treated group than in the scopolamine-treated group ($P < 0.05$). In an *in vitro* test, nodakenin inhibited AChE activity in a dose-dependent manner (IC_{50} 84.7 μ M), consistent with an earlier report of 68 μ M (Kang *et al.*, 2001). In addition, nodakenin was also found to inhibit AChE activity for 6 h in an *ex vivo* test. These results suggest that nodakenin may contribute to alleviation of cognitive impairment, and that its beneficial effects are mediated, in part, via the enhancement of cholinergic signaling.

Combination of AGN with Other Natural Products

Additional studies were reported for multi-herbal composition with AGN on memory (Kang *et al.*, 2005; Kim *et al.*, 2010) and even combined with silkworm (Guo *et al.*, 2021).

Human Trial Status

A phase II clinical trial in elderly Korean patients with memory impairment (Alzheimer's dementia) had been published in Korean language (Kim *et al.*, 2003) with promising efficacy potential. However, the results of a follow-up trial comparing AGN to donepezil, an AChE inhibitor drug, have not been reported (Clinicaltrials.gov NCT01245530). Additional human trials are warranted to optimize AGN dosages.

Prophylaxis and Treatment of Stroke and Thromboembolism

Decursinol Hybrid Drugs

Given the analgesic and neuroprotective activities of AGN and pyranocoumarins, a Korean group synthesized 2 DOH hybrid compounds (see Fig. 3) and tested them against experimentally-induced transient cerebral ischemia (stroke model) in comparison with DOH or the respective drug counterpart (Lee *et al.*, 2012; Yan *et al.*, 2013). In the first study, the α -lipoic acid (ALA) and DOH hybrid (Lee *et al.*, 2012) was compared by oral administrations for neuroprotective effects with ALA, or DOH alone against ischemia–reperfusion damage in the gerbil hippocampal (*cornu ammonis*) CA1 region induced by 5 min of transient global cerebral ischemia via bilateral occlusion of the common carotid arteries (BOCCA). The 20 mg/kg ALA-DOH hybrid pre-treatment regimen (on day -2 , -1 and

30 min before 5-min ischemia) protected the pyramidal neurons from ischemic damage in the CA1 region. In addition, 20 mg/kg ALA-DOH pre-treatments markedly decreased the activation of astrocytes and microglia in the same brain region 4 days after ischemic injury. All the single drugs at 10 and 20 mg/kg and the 10 mg/kg ALA-DOH pre-treatment-ischemia-group showed no neuroprotective effect against ischemic damage 4 days after BOCCA. On the other hand, post-treatments (30 min after 5-min BOCCA plus an additional dose on day +1) with all the same dosages of each drug alone or the hybrid did not show any neuroprotective effect against ischemic damage, supporting only a prophylactic utility of the threshold efficacious dose of the new hybrid drug.

With the same gerbil model for transient global cerebral ischemic damage (BOCCA for 5 min) and Sprague-Dawley rats for transient focal cerebral ischemic damage (90 min middle cerebral artery occlusion, MCAO), the second study investigated neuroprotective effects of aspirin (acetyl salicylic acid ASA), DOH and DOH-aspirin hybrid (DOH-ASA) by *i.p.* injection as either pre-treatment or post-treatment regimen. While 20 mg/kg, not 10 mg/kg, DOH-ASA pretreatments protected against ischemia-induced neuronal death after transient focal and global ischemic damage, each of the single drugs alone at either dose as pretreatments failed to show a neuroprotective effect. In addition, 20 mg/kg DOH-ASA pre-treatment reduced the ischemia-induced gliosis and maintained antioxidant levels in the corresponding injury regions. Like the DOH-lipoic acid hybrid in the first study, the DOH-ASA hybrid or each drug alone given as post-treatment failed to show any therapeutic effect. Thus, the DOH-ASA hybrid at 20 mg/kg threshold dose was only of a prophylactic utility to protect neurons from ischemic damage, and attenuated ischemia-induced gliosis and maintained anti-oxidant enzyme protection, as did the DOH-ALA hybrid drug.

Notably, the hybrid drugs showed demonstrable prophylaxis efficacy when DOH or the respective counterpart drug failed to protect against ischemia stroke at the same threshold dosage (20 mg/kg); this was evident in spite of a bias against the hybrids on molar dosage basis. In these hybrid drugs, one might speculate that DOH might have acted as a neuron/brain drug carrier to enhance delivery of lipoic acid and aspirin payloads or that the hybrids represented novel drug entities with distinct functions from the components. Tissue uptake and metabolite tracking studies with these hybrid drugs vs. component entities are warranted to address these hypotheses.

AGN Alcoholic Extracts on Ischemia Stroke Models

Very recently, two groups reported the impact of alcoholic extracts of AGN on MCAO-induced stroke in mice (Lee *et al.*, 2020b; 2021a) and the global cerebral ischemia model by bilateral common carotid artery occlusion (BCCAO) in gerbils (Lee *et al.*, 2021b). In the mouse work in the prophylactic setting (Lee *et al.*, 2020b), a methanol (99%) extract of AGN root was orally administered once (–1 h) or twice (–24 h and –1 h) prior to starting a 90-min MCAO. The two-dose AGN pre-treatment regimen at 1000 mg/kg reduced the infarct volumes in ischemic brain by ~ 40% and ameliorated

the morphological deteriorations in the brain neuron cells, whereas the acute single pre-treatment did not offer protection. The same group expanded the study by modifying the MCAO time to 2 h and determined the dose response in the prophylactic setting with 300, 1000 and 3000 mg/kg as acute pretreatment (–1 h) or given on two consecutive days (–24 h and –1 h) (Lee *et al.*, 2021a). They also included the 1000 mg/kg dose as a post-ischemia therapy in comparison with nimodipine, all given 1 h after MCAO. While the single dose acute pre-treatment (–1 h) with all three doses of AGN failed to decrease infarct volume, the two-dose regimen at 1000 mg/kg decreased the infarct volume from 50% to 15%, whereas neither the 300 mg/kg nor the 3000 mg/kg dosage showed a reduction (i.e., a bell-shaped dose response curve). The post-ischemia treatment with 1000 mg/kg AGN or nimodipine did not show any therapeutic benefit. In either study, the 1000 mg/kg prophylactic two-dose regimen ameliorated brain edema but not the neurological deficit score from the MCAO. Biomarker analyses associated the protective two-dose AGN regimen with decreased neuronal death and changes of ERK-related signaling and PI3K/AKT/mTOR (but not JNK or P38MAPK) pathways and inflammatory cytokine (TNF- α , IL-6) expression in the ipsilateral hippocampus hemisphere (Lee *et al.*, 2020b; 2021a).

Together, these two mouse studies indicated that sustained prophylactic use of AGN in an optimal dose range may help to prevent or lessen ischemia damage. Whether the protective activity was mediated by pyranocoumarins and the merit for post ischemia therapy would require additional investigation.

The “therapeutic utility” question was addressed in a gerbil global cerebral ischemia model with bilateral occlusion of the common carotid arteries (BOCCA) (Lee *et al.*, 2021b). Thirty minutes following completion of 5-min BOCCA, the investigators treated the inflicted gerbils orally with AGN at 200, 350 and 400 mg/kg dosages (extracted with 98% ethanol) to define the dose response and 25 mg/kg of D corresponding to an efficacious dosage of AGN at 350 mg/kg, as their AGN extract contained a measured D content of 7.3%. They showed that the pyramidal neurons located in the *cornu ammonis* 1 (CA1) among the hippocampal subfields were dead at 5 days after ischemia episode; however, treatment with the AGN extract (350 and 400 mg, not 200 mg) and D preserved the pyramidal neurons after ischemia. The ischemia caused a breach of blood–brain barrier (BBB) as detected by immunoglobulin G (IgG) extravasation in CA1 parenchyma from 2 days after the BOCCA, but BBB/IgG leakage was dramatically attenuated in the CA1 parenchyma treated with the AGN extract and D. Furthermore, astrocyte endfeet, as a component of the BBB, were severely damaged at 5 days after the ischemia; however, post-treatment with AGN extract and D dramatically attenuated the damage to the endfeet. The post-treatment with either AGN or D preserved spatial memory in an 8-arm radial maze test (8-ARMT) and learning memory in passive avoidance test (PAT) while the vehicle treated ischemia gerbil sustained memory impairment after 5 days following BOCCA. In sum, therapeutic application of AGN extract after 5 min transient forebrain ischemia protected hippocampal neurons from the ischemia, attenuated ischemia-induced BBB leakage and damage of astrocyte endfeet, and preserved spatial and learning memory, and D recapitulated the treatment outcome.

AGN Water Extract

An AGN root water extract has been also shown to mitigate the course of experimental stroke model by transient focal cerebral ischemia in male Sprague-Dawley rats (Oh *et al.*, 2015). After focal cerebral ischemia was induced with MCAO for 90 min, the rats were given an oral administration of the water extract of AGN. After 24 h reperfusion, the AGN water extract-treated rats significantly decreased the volumes of brain infarction and edema. AGN water extract attenuated the increased BBB permeability (Evans blue leak), and ischemic neuronal death and inhibited the activation of astrocytes and microglia in affected brain region. AGN water extract also significantly increased the expression of angiogenesis and BBB barrier proteins Ang-1, Tie-2, VEGF, ZO-1 and occludin in association with an activation of the PI3K/Akt pathway. AGN water extract also significantly increased the expression of ICAM-1 in ischemic brains.

In summary, the studies with the AGN alcoholic extract showed efficacy in attenuating ischemia–reperfusion stroke models in mice as prophylactic prevention with a bell-shaped dose response (Lee *et al.*, 2020b; 2021a) and as a post-stroke therapy in gerbils and D as a principal active compound (Lee *et al.*, 2021b). However, DOH given alone at reasonably high doses in either a prophylactic or post-ischemia therapy scenario in the two drug-hybrid studies did not show a protective effect by another group (Lee *et al.*, 2012; Yan *et al.*, 2013). Given the hepatic first pass breakdown of D/DA through gastrointestinal route of delivery, it is questionable but may still be conceivable that a sufficient amount of D/DA might be enriched in the ischemic brain to render the protective benefit. In yet to be published work, we have observed that mice having been dosed with DOH survived much longer during CO₂-inhalation euthanasia, which could be interpreted in light of the ischemia ameliorating activity observed in these models. Head-to-head comparison of D/DA vs. DOH in PK-adjusted equivalent dosages will be necessary to settle the *in vivo* active compound issue.

Since the water extract of AGN root was shown to be efficacious as a post-MCAO treatment in the rat model (Oh *et al.*, 2015) and unlikely to contain any D/DA, the active compounds in the water extract could be anti-inflammatory/immune regulatory as reflected in the cytokine normalization outcome.

Epinephrine-Collagen Thromboembolism Model

An AGN ethanolic extract and D (12.5 mg/kg) were evaluated for activity against thromboembolism in an experimentally-induced platelet aggregation model (Bravo *et al.*, 2021). AGN given orally at 40, 80 and 160 mg/kg 1 h before epinephrine and collagen tail vein *i.v.* injection showed a significant reduction in death or paralysis in mice caused by the collagen/epinephrine-induced thromboembolism in a bell-shaped manner, such that the highest AGN dose had less protection than the middle dose, which was more potent than aspirin. There was a corresponding inhibition of collagen-induced human platelet aggregation in a concentration-dependent manner. Additionally, AGN-treated mice did not show severe gastric ulcerating bleeding compared to aspirin-treated mice. Given the D/

DA conversion to DOH following oral route of AGN delivery, a comparison of the efficacy of PK-adjusted equivalent DOH dosage with D and AGN will inform in this pulmonary thromboembolism-inflicted paralysis and death model the issue of *in vivo* active compound and interactions among pyranocoumarins and other AGN phytochemicals to account for the bell-shaped protection response.

Human Trial Status

No human trial has been reported or registered to assess the utility of AGN or its pyranocoumarins for preventing stroke or alleviation of stroke symptoms.

Alleviation of Seizures

Epilepsy is a neurological disorder with recurrent unprovoked seizures as the main symptom. A paper reported the effects of D on a kainic acid (KA)-induced *status epilepticus* model in mice (Lee *et al.*, 2014b). The authors administered D by *i.p.* injections (20 mg/kg) to male 7-week-old C57BL/6 mice 30 min prior to KA (30 mg/kg, *i.p.*), then examined behavioral seizure score, electroencephalogram, seizure-related expressed protein levels, neuronal cell loss, neurodegeneration, and astrogliosis. The D-treated (30 min prior) KA-injected group showed significantly decreased behavioral seizure activity and abated intense and high-frequency seizure discharges in the parietal cortex during 2 h observation compared with the group treated only with KA. Furthermore, their *in vivo* results indicated that D strongly inhibited selective neuronal death, astrogliosis, and oxidative stress induced by KA administration. Therefore, D was able to attenuate KA-induced seizures and could have potential as a prophylactic anti-epileptic drug. Based on metabolism knowledge of D and DA, DOH should be compared side-by-side with these parental drugs to define any role of DOH in the anti-epileptic activity. No human trial has been reported or registered to assess the utility of AGN or its pyranocoumarins for preventing or managing epilepsy.

Sleep Disorder

A recent study tested whether DA augmented pentobarbital-induced sleeping behaviors and the role of GABA_A-ergic systems in rodent models (mice and rats) (Woo *et al.*, 2017). Oral administration of DA (10, 25 and 50 mg/kg) markedly suppressed spontaneous locomotor activity (our interpretation: sedation) for the mice. DA also prolonged sleeping time and decreased the sleep latency by pentobarbital (42 mg/kg), in a dose-dependent manner, similar to muscimol, both at the hypnotic (42 mg/kg) and sub-hypnotic (28 mg/kg) dosages. Especially, DA increased the number of sleeping mice in the sub-hypnotic dosage. In rats with implanted EEG monitoring, DA (50 mg/kg, *p.o.*) modulated sleep architectures and reduced the counts of sleep/wake cycles. At the same time, DA increased total sleep time, but not non-rapid eye movement (NREM) and rapid eye movement (REM) sleep. At the molecular/receptor level, DA (0.001, 0.01 and 0.1 µg/ml) increased intracellular Cl⁻ influx

level in cultured hypothalamic primary neuronal cells of rats. In addition, DA increased the protein expression of glutamic acid decarboxylase ($GAD_{65/67}$) and $GABA_A$ receptors subtypes. Taken together, these results were interpreted to support that DA potentiated pentobarbital-induced sleeping behaviors through the activation of $GABA_A$ -ergic systems and could be useful in the treatment of insomnia for humans. Based on metabolism and PK knowledge of D and DA, DOH might have contributed to improve sleep (sedation; locomotor decline) and should be tested side-by-side to address this prediction.

No human trial has been reported or registered to assess the utility of AGN or its pyranocoumarins for managing or alleviation of sleep disorders.

Anxiety Management

Stress and depression especially during a pandemic crisis like COVID is a significant public health and clinical management issue. It was not surprising to identify an anxiolytic effect of AGN due to its many neuroprotective attributes.

Corticosterone Stress Anxiety Model

Lee *et al.* (2015) examined the effects of AGN (methanol 80%) extract treatment in a rat model of depressive and anxiety-like behaviors, induced by chronic stress hormone corticosterone exposure. Repeated injection of corticosterone induced dysregulation in the hypothalamic–pituitary–adrenal (HPA) axis, resulting in depression. Activation of the HPA axis in response to the repeated corticosterone injections was confirmed by measuring serum levels of corticosterone and the expression of corticotropin-releasing factor in the hypothalamus. In their experiment, male rats received 10, 20, or 50 mg/kg AGN by *i.p.* injection daily at 30 min prior to a daily injection of corticosterone for 21 consecutive days. The results showed that daily AGN administration significantly reversed the depression and anxiety-like behavioral abnormalities in a dose-dependent manner. It also blocked increases in tyrosine hydroxylase expression (dopamine synthesis) in the *locus coeruleus* and rescued the decreased expression levels of brain-derived neurotrophic factor (BDNF) and its receptor TrkB mRNAs in the hippocampus. When compared with the antidepressant drug fluoxetine (Prozac) (a selective serotonin reuptake inhibitor), AGN extract was fully efficacious but 1/5 as potent (50 mg AGN/kg $\approx$ 10 mg/kg fluoxetine) at regulating the above behavioral and biochemical metrics. These findings indicate that administration of AGN prior to high-dose exogenous corticosterone significantly improved depressive behaviors, possibly by modulating the central noradrenergic system and regulation of BDNF expression in rats.

Serotonin Level Increased by DOH

Along the line of SSRI, DOH gavage has been found to increase the serotonin (5-hydroxytryptamine) level in rat brain (Abdullayeva *et al.*, 2009). The highest concentration

of serotonin was observed in frontal cortex and striatum after administration of 10 mg DOH/kg body weight.

Human Trial Status

To our best knowledge, no human trial has been reported or registered to assess the utility of AGN for the managing or alleviation of psychiatric disorders associated with depression and anxiety disorders.

Inflammation and Inflammatory Diseases

Cytokine storm as our body's response to infection and inflammation can lead to serious side effects and negative health consequences, including death. Reports in different animal models of inflammatory diseases and conditions appeared to tie AGN extracts with dampening cytokine expressions and their signaling to inflammatory disease prophylaxis and management.

Acute Inflammation/Allergy Models

Anti-inflammatory effects of an AGN ethanol extract had been evaluated more than a decade ago in animal models with air pouch (Shin *et al.*, 2009) and ear edema (Joo *et al.*, 2010). In the air pouch model (Shin *et al.*, 2009), an ethanol extract of AGN (50, 160, or 500 mg/kg, oral dosing) was tested against carrageenan (a linear sulfated polysaccharide extracted from seaweeds)-induced air pouch inflammation model by an *s.c.* injection of 1 ml of carrageenan (1%) under dorsal skin of mice. The carrageenan-loaded air pouches markedly increased the exudate volume and exudate albumin concentration, which were significantly attenuated by oral pretreatment with AGN extract. AGN extract also markedly reduced carrageenan-induced infiltrations of neutrophils, monocytes, and lymphocytes but did not influence eosinophils or basophils. Carrageenan dramatically increased levels of TNF- α and IL-6, elevated nitric oxide, and slightly increased prostaglandin E2 (PGE2). AGN extract pretreatment significantly attenuated the rise of TNF- α , and nitric oxide, but did not alter the response of IL-6 or PGE2.

The same lab studied the anti-allergic effects of AGN extract by following ear erythema, ear weight, vascular leakage, hematology, TNF- α , IL-6 and immunoglobulin E (IgE) from mice sensitized with 2,4-dinitrofluorobenzene (Joo *et al.*, 2010). The authors found that oral administration of AGN at 50 to 500 mg/kg dosages dissolved in 50% ethanol was able to reduce ear erythema and ear weight, suppress Evan's blue leakage from vasculature and decrease circulating levels of IL-6 and IgE. These results indicate that AGN extract attenuated the TNF- α -nitric oxide pathway or IL-6 inflammatory cytokine responses, specific to the inflammation models.

A recent study examined the effect of AGN gavage on allergic rhinitis induced by ovalbumin (OVA)-sensitization and nasal challenge model in mice (Ji *et al.*, 2021). Following

3 weekly *i.p.* injections of 2 mg aluminum hydroxide and 50 µg OVA, the mice were challenged with 4 intranasal injections of 400 µg OVA every other day starting on day 21. AGN extract prepared in 70% ethanol (final yield 39%) was administered by gavage daily for 1 week from the start of OVA challenge at 30 and 100 mg/kg dosages. Dexamethasone (Dex, 1 mg/kg) gavage served as a positive anti-inflammation steroid control. AGN gavage exerted a dose-dependent reduction of the OVA-induced rubbing and sneezing allergic rhinitis behavior, improved blood immune cells, histopathological and serum cytokine parameters, albeit less potent than Dex.

Chronic Injury Inflammation Models

In a chemically-induced inflammation and fibrosis model (Choi *et al.*, 2014), D was shown to attenuate carbon tetrachloride (CCl₄)-induced liver fibrosis. Liver injury was induced *in vivo* by *i.p.* injection of CCl₄ with or without D for 4 weeks in mice. D treatment significantly reduced the ratio of liver/body weight, α -smooth muscle actin (α -SMA) and collagen I overexpression in CCl₄ treated mice liver. D treatment blunted the elevated serum levels of liver damage marker enzymes, including ALT, AST, and ALP. D effectively blocked the levels of NAD(P)H oxidase (NOX) protein, and Smad 2/3 phosphorylation in liver of mice exposed to CCl₄. Using cell culture of a human immortalized hepatic stellate cell line, the authors showed that D markedly attenuated the generation of reactive oxygen species, NOX protein (1, 2, and 4) upregulation, NOX activity, and superoxide anion production induced by TGF- β 1. It also significantly reduced TGF- β 1-induced Smad 2/3 phosphorylation, nuclear translocation of Smad 4, and association of Smad 2/3-Smad 4 complex. Given the liver first pass metabolism of D and DA, the exposure of hepatic cells to these parent pyranocoumarins from an AGN supplement could be substantial and relevant as a liver protectant.

Inflammatory Bowel Disease-Ulcerative Colitis (IBD-UC) Model

Recently, AGN ethanol extract (95% ethanol) was reported to attenuate the disease course of a chemically-induced IBD-UC model in mice (Oh *et al.*, 2017). In the dextran sulfate sodium (DSS, 5% in drinking water)-induced IBD-UC model, disease activity index values were measured by clinical signs such as a weight loss, stool consistency, rectal bleeding and colon length. Daily oral treatment with the AGN at 10, 20, and 40 mg/kg for 7 days alleviated weight loss, decreased disease activity index scores, and reduced colon shortening in mice with DSS-induced UC. Hematoxylin and eosin staining of the colon showed significant alleviation of the tissue damage by AGN. AGN dosing inhibited the production of IL-6 and TNF- α inflammatory cytokines (probably from macrophages) in serum and colon tissue. Moreover, AGN suppressed the increased expression of COX-2 and HIF-1 α and the increased production of protective PGE₂ in colon tissue.

A different group studied DA in the same IBD-UC model and reported a prophylactic/therapeutic potential of DA for managing UC (Thapa *et al.*, 2019). DA per *i.p.* injection treatment every other day (0.4 and 4 mg/kg starting day -1 before DSS (2.5% in drinking

water) attenuated the severity of colitis including a reduction in weight loss, colon shortening dose-dependently, and a protection from colonic tissue damage induced by DSS only at the high dose. The efficacious dose of DA treatment decreased type 17 helper T (Th17) cells and neutrophils in the colitis tissues. They performed cell culture studies to demonstrate a suppressive effect of DA on Th17 cell differentiation and IL-17 production without affecting the survival and proliferation of CD4 T cells. Even though this study substantiated DA as a contributor to DSS-UC attenuation, how much of Th17 modulation *in vivo* was attributed to DA or its metabolite DOH remains to be determined. Overall, these studies in DSS-UC model and documentation of anti-inflammatory and modulatory effects on inflammatory T cells suggest the possibility of prevention and clinical management for IBD.

Human Trial Status

No human trial has been reported or registered to assess the utility of AGN or its pyranocoumarins for IBD treatment or management or the pain associated with IBD.

Sepsis

Prophylaxis and Treatment Effects of DOH Against Sepsis Models

An early study examined the effect of DOH against three models of sepsis using a small number of mice per group ($n = 5$) (Jung *et al.*, 2008). In a prophylaxis setting, DOH was injected *i.p.* to pretreat mice with various doses (1–100 mg/kg) for 30 min before induction of experimental sepsis in ICR mice, i.e., lipopolysaccharide (LPS)/D-galactosamine (GalN) or high-dose LPS (20 mg/kg) by *i.p.* injections. In each of the LPS models, there was a dose-dependent reduction of sepsis lethality ($ED_{50} \sim 10$ mg/kg) with complete prevention of death at the highest dose. Intra-peritoneal pretreatment of mice with DOH (50 mg/kg) markedly enhanced the LPS/GalN-induced increase of plasma IL-10 levels, without affecting plasma TNF- α , IL-6 and IL-12 levels.

In a “therapy” setting, DOH was injected *i.p.* at up to 50 mg/kg twice at 2 h and 4 h after cecal ligation and puncture (CLP) to induce sepsis. The post-CLP model showed a seemingly plateau level of therapeutic benefit at 10 mg/kg. In spite of the small number of mice each group, these results suggested DOH might be an active compound for the prophylaxis of sepsis and might even be therapeutic during the early onset stage of sepsis.

MRSA-Infection Sepsis Model

In a recently published study of a bacterial infection sepsis model (Pak *et al.*, 2021), DA was tested in mice infected with a lethal dose of methicillin-resistant *Staphylococcus aureus* (MRSA). DA pre-treatment (*i.p.* every other day for three doses, 0.4, 2 and 10 mg/kg) decreased the mortality dose-dependently, reversed the bacteremia, and attenuated the cytokine storm, which was associated with decreased CD38(+) macrophage populations in the blood and liver. *In vitro* studies revealed that DA inhibited the functional activation

of macrophages in the expression of pro-inflammatory mediators in response to microbial infection, while promoting the bacterial killing ability with an increased production of reactive oxygen species. Mechanistically, DA treatment attenuated the NF-kappaB and Akt signaling pathways. However, given the earlier observed anti-sepsis action of DOH discussed above (Jung *et al.*, 2008) and in light of the first pass conversion of DA to DOH, such *in vitro* mechanisms may be moot.

Decursinol Derivatives

Given antiseptic activity of DOH, Lee *et al.* synthesized D/DA-analog derivatives (KC1-3) (see chemical structures in Fig. 3) and evaluated their activities on high mobility group box 1 (HMGB1)-mediated septic responses and survival rate in a mouse model of CLP-induced sepsis. KC1 and KC3, but not KC2, significantly reduced HMGB1 release in LPS-activated human umbilical vein endothelial cells (HUVECs) and attenuated the CLP-induced release of HMGB1. The *in vivo* analyses revealed that KC1 and KC3 reduced sepsis-related mortality and tissue injury.

The same group found that JH-4 (see Fig. 3), with a structure similar to KC1 (without the amide linker), also attenuated HMGB1-mediated septic responses and improved the survival rate in the CLP-induced mouse sepsis model (Lee *et al.*, 2019c). With these chemical modifications, these derivatives were more potent than DOH to improve survival of the mice from sepsis (ED50 ~ 0.58 mg/kg by *i.p.* injection for KC1, KC3 and JH-4 vs. ED50 ~ 10 mg DOH/kg). The structure–activity relationship (SAR) of these 4 analogs (3 active, 1 inactive) warrants in-depth follow up on the determinants of differential activity outcomes based on only subtle variations on the side chain structure.

Human Trial Status

No human trial has been reported or registered to assess the utility of AGN or its pyranocoumarins for preventing or treat sepsis.

Metabolic Syndromes

The health problems associated with over-nutrition and nutrient imbalances in the USA have reached epidemic levels. These problems are driven by too high sugar and fat consumption and too little physical activity, manifesting as hyperglycemia, diabetes, overweight/obesity. Studies in animal models, especially more recent publications, have supported a potential for AGN or its phytochemicals as a dietary supplement to regulate glucose homeostasis and lipid metabolism.

High Fat-Diet Induced Obesity Animal Models

An earlier study (Hwang *et al.*, 2012) investigated the anti-obesity potential of D using mice fed a high-fat diet (HFD) and a HFD plus D 200 mg/kg diet (HFD + D) in comparison to

a normal diet (ND) for 7 weeks. Mice administered HFD + D showed a decrease in weight gain, triglyceride content, total cholesterol content and fat size compared with those that received the HFD alone; this was observed despite similar quantities of total food intake. Furthermore, D improved glucose tolerance in mice fed a HFD. Finally, administration of D along with the HFD significantly reduced the secretion of HFD-induced adipocytokines such as leptin, resistin, IL-6 and MCP-1. Mechanistically, D was tested for ability to inhibit adipocyte differentiation in mouse 3T3-L1 cells *in vitro*. Decursin treatment resulted in the inhibition of adipocyte differentiation and the expression of fatty acid synthase.

Diabetes

The antidiabetic and lipid metabolism effect of AGN extract (AGNE) was evaluated in a type 2 diabetic mouse model (Bae *et al.*, 2016a). Male C57BL/KsJ-db/db mice were treated by dietary supplementation with either AGNE (10, 20, and 40 mg/kg or ppm) or metformin (100 mg/kg, or ppm) for 8 weeks. AGNE dietary supplementation (20 and 40 mg/kg, ppm) significantly decreased fasting glucose and insulin levels, decreased the areas under the curve of glucose in oral glucose tolerance and insulin tolerance tests, and improved homeostatic model assessment-insulin resistant (HOMA-IR) scores. AGNE also ameliorated hepatic steatosis, hyperlipidemia, and hypercholesterolemia. Mechanistic studies suggested that the glucose-lowering effect of AGNE was associated with the activation of AMP-activated protein kinase (AMPK), Akt, and glycogen synthase kinase-3. In many of these metrics, AGNE was five times more potent than metformin in regulating the beneficial biochemistry and physiology.

The same group explored the impact of AGN on lipid dysregulation in more detail (Bae *et al.*, 2017). In this study, C57BL/6J mice were fed a HFD, or HFD with various doses of AGNE for 16 weeks. Supplementation with AGNE-attenuated glucose and insulin intolerance, hepatic steatosis and inflammation, and hypertriglyceridemia induced by the HFD. AGNE significantly suppressed hepatic *de novo* lipogenesis in association with an activation of AMPK. In cell culture model, HepG2 cells treated with free fatty acid mixture (oleate:palmitate = 2:1) displayed increases of *de novo* lipogenesis and consequent lipid droplet formation. Addition of D or DA increased Sirt1 expression, which suppressed lipid accumulation in HepG2 cells. These results indicate that the metabolic effects of AGNE may be related to the induction of Sirt1 and consequent activation of AMPK.

Yet another study reported cell culture results of stem cells exposed to D or DA (Park *et al.*, 2019). As visceral adiposity is closely associated with metabolic disorders and cardiovascular diseases, the authors investigated adipose-derived stem cells (ASCs) isolated from visceral adipose tissue (VAT). Among four coumarin compounds of AGN, D and DA significantly inhibited adipocyte differentiation from ASCs. D and DA downregulated CCAAT/enhancer binding protein α (C/EBP α), peroxisome proliferator-activated receptor γ (PPAR γ), adipocyte fatty acid binding protein (aP2), fatty acid synthase (FAS), and acetyl-CoA carboxylase (ACC) at both mRNA and protein levels. Next, treatment with adipogenic differentiation medium (ADM) on ASCs downregulated β -catenin protein expression, while addition of D and DA could restore protein expression and nuclear translocation of

β -catenin suppressed by ADM. D and DA treatment on ADM treated ASCs increased inhibitory phosphorylation of glycogen synthase kinase (GSK)-3 β , thereby preventing β -catenin degradation. Additionally, transfection of *siRNA* against β -catenin significantly upregulated protein expression of C/EBP α and PPAR γ , alleviating the anti-adipogenic effect of D and DA on ADM treated ASCs. Whereas the *in vitro* data overall support D and DA as active compounds to suppress adipogenesis through activation of the β -catenin signaling pathway in ASCs derived from human VAT, it would have been interesting to compare with DOH because of the extensive first pass hepatic metabolism of D and DA (PK/metabolism section). Due to the high lipophilicity of D and DA, perhaps sufficient accumulation still is possible in extrahepatic adipose tissue to modulate the pathways identified in the *in vitro* exposure model.

Anti-Obesity Effect of Furanocoumarin Nodakenin

In addition to pyranocoumarins, the furanocoumarin glucoside nodakenin in AGN was reported to repress obesity and its complications *in vivo* (Jin *et al.*, 2021) in a high fat diet-induced obesity model. Mice were fed a high fat diet (45% fat) for 8 weeks to induce obesity and were given oral nodakenin (10 and 20 mg/kg body weight) for 5 more weeks to compare with orlistat (FDA approved drug to decrease fat absorption) at the same dosages. Nodakenin showed comparable potency to orlistat for suppression of weight gain, dyslipidemia and the development of fatty liver in HFD-induced obese mice normalizing toward mice raised on regular fat diet. In addition, the authors linked inhibition of adipogenic differentiation and obesity-induced inflammation and oxidative stress to the suppression of the VLDLR and MEK/ERK1/2 pathways. In the differentiated 3T3-L1 adipocytes cell culture model, nodakenin regulated adipogenic transcription factors and genes associated with triglyceride synthesis.

Human Clinical Studies

Two human studies have been initiated in Korea (NCT03258229; NCT03079648). The outcome of NCT03079648 has been reported (Jung *et al.*, 2020). In this 12-week clinical trial, with a randomized, double-blind, placebo-controlled design to assess the effects of AGN extract on blood triglycerides, subjects who met the inclusion criterion (130 mg/dL $\leq$ fasting blood TG $\leq$ 200 mg/dL) were recruited. Fifty were assigned to the AGNE group (200 mg/d capsule, equivalent to 1 g/day of AGN) or to placebo control group (capsules with filler) for 12 weeks. Efficacy outcomes (TG, lipid profiles, atherogenic index) and safety parameters were assessed initially for a baseline measurement and after 12 weeks. The results showed after 12 weeks of supplementation, TG and very low-density lipoprotein cholesterol (VLDL-C) concentration and TG/HDL-C ratio in the AGNE group were significantly reduced compared to the placebo group ($p < 0.05$). No significant changes in any safety parameter were observed. These results suggest that the ingestion of AGNE may improve TG and be useful to manage or prevent hypertriglyceridemia. Optimization of dosage and long-term safety should be further pursued.

Osteoporosis and Osteoarthritis

Rat Menopause Model

One study evaluated the effect of particle size of AGN root powder on its efficacy for treating estrogen-deficit related symptoms of menopause in an ovariectomized (OVX) rat model, starting 5 weeks after surgery (Choi *et al.*, 2012). The AGN roots were pulverized into coarse powder or ultrafine powder and orally administered to the OVX rats at a daily dose of 340 mg/rat (higher than 500 mg/kg body weight in Abstract) for 8 weeks. The ultrafine AGN powder allowed more efficient extraction (5-fold more) of D and DA compared to coarse powder in a simulated absorption test. Compared to sham-operated rats (137 g gain), OVX group gained more weight (181 g), but neither coarse (CRS) (185 g) nor ultrafine (UF) (185 g) AGN group prevented the weight gain. The serum levels of estradiol and progesterone were restored in the OVX_UF group to sham-operated group levels, yet the OVX_CRS group only partially alleviated the drop in estradiol, and did not restore progesterone levels. The OVX_UF group restored LH (luteinizing hormone) to sham-operated level but did not normalize follicle-stimulating hormone (FSH), whereas the OVX_CRS group was not active to restore LH. Serum bone-alkaline phosphatase/total-alkaline phosphatase index reflecting bone regeneration in the OVX_UF group was significantly attenuated from OVX group, whereas the OVX_CRS group experienced little attenuation. In addition, various bone strength metrics in the OVX-Uf group were restored to sham-operated group levels, whereas OVX-CRS has less impact or no impact at all on these metrics. Therefore, ultrafine grinding enhanced the AGN pyranocoumarin bioavailability and nearly restored key female reproductive hormone LH, and bone strength metrics in OVX-rats but did not ameliorate the metabolic changes related to weight gain.

Bone Loss Induced by Bacterial Toxin

In a prophylactic regimen, D blocked lipopolysaccharide (LPS)-induced bone erosion *in vivo* in a mouse model (Kim *et al.*, 2015). One day before the first of two injections of LPS (day 1 and day 4) and subsequently on alternate days throughout a 2-week period, D was administered by *i.p.* injections (2 mg/kg body weight). Dynamic interaction between osteoblastic cells and osteoclastic cells maintains the homeostasis of the bone. The authors explored whether D affected receptor activator of nuclear factor κ B ligand (RANKL)-mediated osteoclastogenesis and reported an efficient inhibition of RANKL-activated osteoclast differentiation by inhibiting transcriptional and translational expression of nuclear factor of activated T cells (NFATc1), a major downstream target of RANKL. Furthermore, D decreased fusion and migration of pre-osteoclasts by downregulating mRNA expression levels of DC-STAMP and β 3 integrin, respectively. D therefore inhibited osteoclastogenesis and inflammatory bone loss via blockage of NFATc1 activity and fusion and migration of pre-osteoclasts.

A different lab reported D reduced LPS- or OVX-induced bone loss *in vivo* (Wang *et al.*, 2016) and similar mechanistic findings by investigating the effects of D on

RANKL-induced osteoclast formation using mouse bone marrow-derived macrophages (BMMs). D inhibited RANKL-induced osteoclast formation without cytotoxicity. In particular, D maintains the characteristics of macrophages by blocking osteoclast differentiation by RANKL. Furthermore, *in vivo* LPS/RANKL-stimulated and OVX-induced bone resorption was diminished by D at *i.p.* dosage of 20 mg/kg daily injection in therapeutic setting. Mechanistically, D blocked the RANKL-triggered ERK mitogen-activated protein kinases (MAPK) phosphorylation, which results in suppression of c-Fos and NFATc1 expression. D could be useful for the treatment of bone diseases associated with excessive bone resorption. DOH should be studied to address whether it accounts for the therapeutic benefit of D.

Osteoarthritis

In a recent study of the osteoarthritis model, D displayed significant treatment activity in mouse (He *et al.*, 2021). Surgically-induced osteoarthritis was achieved by destabilization of medial meniscus (DMM) in male C57B mice. After the surgery, D treatment (daily intragastric gavage, 20 mg/kg, 8 weeks) attenuated the destruction of articular cartilage (severity score and subchondral cortical bone thickness) and decreased the serum inflammatory factor (IL-1 β , IL-6 and TNF- α) levels to below the sham-operated control levels. *In vitro* cell experiments with primary isolated chondrocytes showed that the inflammatory response in chondrocytes was mediated via IL-1 β , which led to abnormal secretion of pro-inflammatory factors, such as prostaglandin E2 (PGE2), IL-6, TNF- α , cyclooxygenase-2 (COX-2), nitric oxide (NO) and inducible nitric oxide synthase (iNOS). These cytokines were all decreased by preconditioning with D in a concentration-dependent manner from 1 to 10 μ M. Moreover, D could restrain IL-1 β -mediated inflammatory reaction and the collapse of extracellular matrix (ECM) via reducing the secretion of a disintegrin and metalloproteinase with thrombospondin motifs (ADAMTS) and matrix metalloproteinases (MMPs). In short, D restrained IL-1 β -mediated abnormal excitation of PI3K/AKT/NF- κ B axis. Furthermore, molecular docking analysis suggested binding of D with the inhibitory targets of PI3K. It is remarkable that primary chondrocytes viability was improved with lower micromolar D and even 5 μ M was found quite active on these cytokines in cell culture exposures. DOH should be evaluated in the animal model to address whether the active compound is D or its metabolite.

Human Trial Status

Although not a trial for AGN alone, a published study in the US (Chang *et al.*, 2012) had used a double blinded, placebo controlled randomized design to test EstroG-100, a combination of standardized extracts of AGN and *Cynanchum wilfordii*, *Phlomis umbrosa* on menopausal symptoms. The trial was performed for 12 weeks with 64 pre-, peri- and post-menopausal White Hispanic, White non-Hispanic and African American women who were randomly allocated to either the EstroG-100 group ($n = 31$) or the placebo group

($n = 33$). Primary end-points were the mean change in scores of the Kupperman menopause index (KMI) that evaluates 11 symptoms, and the mean change in scores of vaginal dryness. The mean KMI score was significantly reduced in the EstroG-100 group from 29.5 ± 7.4 at baseline to 11.3 ± 5.8 ($p < 0.01$) compared with change of the placebo group (29.2 ± 6.6 at baseline vs. 23.7 ± 7.7 at week 12). The constituting symptoms of vasomotor, paresthesia, insomnia, nervousness, melancholia, vertigo, fatigue and rheumatic pain were significantly improved in the EstroG-100 group in comparison with the placebo group ($p < 0.05$). Statistically significant improvement in vaginal dryness in the EstroG-100 group was also observed compared with that of the placebo group ($p < 0.05$) and without weight gain or any serious side effects.

Spermatogenesis/Male Fertility

Cryptorchidism is a condition in which one or both of the testes fail to descend from the abdomen into the scrotum, leading to lower or no viable sperms due to hyperthermic damage inside the body than in the scrotum. Higher testicular temperature results in altered spermatogenesis due to heat-related oxidative stress. One paper reported an examination of AGN effects on a cryptorchidism-induced infertility rat model and on antioxidant activity *in vitro* in TM3 Leydig cells (Bae *et al.*, 2016b). In the rat model, either distilled water or 400 mg/kg of AGN extract (prepared in 30% ethanol) was administered orally for 1, 2 and 4 weeks after unilateral cryptorchidism was performed. At each time point, six rats from the control group and six rats from AGN treatment group were sacrificed. Testicular weight, semen quality, anti-oxidant activities, nuclear factor erythroid 2-related factor 2 (Nrf2) protein, and mRNA expression of Nrf2-regulated genes were analyzed. Treatment with AGN extract improved the mean weight of the cryptorchid testis, maintained sperm counts, motility, and spermatogenic cell density, decreased levels of 8-hydroxy-2-deoxyguanosine (8-OHdG) and increased levels of superoxide dismutase (SOD), significantly increased Nrf2 and heme oxygenase-1 (HO-1), and significantly increased protection against apoptosis ($\uparrow$ Bcl/ $\downarrow$ Bax). In the cell culture model, the AGN extract protected TM3 Leydig cells against hydrogen peroxide (oxidative stress)-induced death. Although these authors attributed the beneficial activities to D in the publication title, the compound was not tested against either animal model or cell culture.

No human trial has been reported or registered to assess the utility of AGN or its pyranocoumarins for fertility benefits.

Human Clinical Translation Challenges

AGN Herbal Products

The source of raw materials and standardization of AGN extraction procedures as well as the formulation and dosing regimens for future studies will remain major challenges for cancer indication and in other diseases. These issues are not unique to AGN studies, as all

herbal/botanical investigations face the same problems of quality control (QC) and quality assurance (QA) for product consistency. An herbal industry-wide consensus for horticulture and harvest, post-harvest processing and storage to ensure relatively uniform raw source materials, standardization of protocols of extraction and manufacturing for AGN extracts and QC/QA practices will be desirable and warranted.

Optimizing Trial Design

As reviewed above, animal models have indicated many potential health benefits from AGN whole root formulations, varieties of alcoholic extracts and water extracts, whereas few have been clinically tested in human subjects or patients with the desired rigor. Properly powered, randomized, blinded and placebo-controlled trials are the best approaches. However, choosing the optimal dosages for testing will be crucial, as insufficient dosage to deliver the “active” principles of any given extract type will lead to false-negative outcomes that may pre-maturely short-change a beneficial indication if properly tested at the correct dosages. We recommend Phase I dose-escalation trials to determine Phase II recommended doses for each type of extract in the intended target subject/patient populations prior to efficacy testing in Phase II trials. Such staged progression of trials ensures that the efficacy testing conditions in Phase II trials are maximized for positive signals.

Future Research Directions

Human Clinical Trials

In spite of publication bias over-representing “positive” outcomes of animal disease model studies, the *in vivo* bioactivities of AGN extracts or their phytochemicals reviewed above indicate a multitude of potential translatable health benefits (prevention as well as treatment) to humans, provided that they are delivered in sufficient and safe dosages and duration of exposure and balanced out for risk of adverse side effects/toxicity. Therefore, to reiterate a crucial point for future human clinical trials, as alluded to in the previous section, they should be designed in stages, that is, Phase I trials in target subject populations and disease patient cohorts to study dose safety information, both acute and chronic intake, to establish maximally tolerated dose (MTD) as well as Phase II recommended dosage (P2RD). Innovative Bayesian adaptive Phase I/II trial designs such as EffTox could help to optimize dose selection (Yan *et al.*, 2018). Once dose safety data are on hand, proceed to Phase II efficacy trials, preferably with double-blinded, randomized, placebo-controlled trials with sufficient statistical power. Reliably consistent herbal supplement test articles, preferably produced with standardized extraction protocols and manufacturing controls from horticulturally quality ensured raw source materials, will be crucial to ensure the reproducibility and future clinical indications for human use.

For health conditions in which putative active compound(s) is supported by strong animal efficacy testing (as examples, pain killing and sepsis prophylaxis by DOH), clinical trials with an arm for test articles composed of the said principal are warranted for two

reasons. The first is the better pharmaceutical QC/QA for a chemical entity than for herbal extract supplements that will ensure more consistent delivery of the active compound to patients. The second is affirmation of mechanisms of action and the ability to assess the contribution of and interactions with other phytochemicals in the herbal extract supplements to the health benefit. Potential interactions with the other components could be positive, neutral or negative for the uptake, metabolism and distribution of the active compound or modulating its adverse effects.

For alcoholic extracts containing D and DA, the major gaps of knowledge for PK and metabolism in humans as well as in animal models are the nature of the exposure dose-response and whether metabolic ceilings exist that may limit the proportion of D and DA converting to DOH as the intake increases. Although not systematically investigated, available structure-activity knowledge of the parent compounds and metabolite in cell culture models in terms of cell cycle arrest, cell death and AR or estrogen receptor signaling by us (Jiang *et al.*, 2006; Guo *et al.*, 2007; Jiang *et al.*, 2007) and others (Yim *et al.*, 2005) supports very different effects of DOH from D and DA, therefore changing their relative circulating or target tissue levels through dosage manipulation or through drug–drug interactions that alter their metabolism as we have demonstrated with CYP inhibitors (Zhang *et al.*, 2015b) could shift PD targets and mechanisms of action. Such novel knowledge will be exportable among subjects/patients of many health conditions. Similarly, PK and metabolism studies should be expanded to other identified AGN phytochemicals, such as nodakenin and xanthotoxin.

Animal Modeling of “Active Compounds”, PK/PD and Efficacy

As reflected in the preceding review of the models of different health/disease conditions, the understanding of the “active compounds” for many of them are fragmentary and incomplete. Animal models provide the proving ground for identifying the active compounds (prodrug or ultimate) through comparison with the extract type that contains them. For alcoholic extracts, pyranocoumarins have been assumed the *de facto* active compounds for cancer and many of the other bioactivities reported so far. The knowledge to date on PK and metabolism of D and DA and first-pass hepatic conversion to DOH have been well established for current dietary supplement dosage range (See Fig. 4 summary). Animal modeling of PK dose response in acute dose exposure and after repeated dosing will be crucial and needed to inform the metabolic ceiling speculation. As it is not ethically justifiable at present to study tissue uptake and deposition of these and other AGN compounds in human subjects, the animal modeling of tissue uptake and deposition of AGN signature compounds and their metabolites will provide critical information to extrapolate to human biology and likely responses of human organs to elevated and repeated dosages of AGN exposure.

As an example of *in vivo* active compound validation, in a primary carcinogenesis interception study, we have observed D/DA administered orally equimolar to that provided in the AGN extract did not fully account for the inhibitory efficacy of AGN extract on neuroendocrine carcinomas, in spite of equal inhibitory action on the prostate epithelial lesions (Tang *et al.*, 2015). Therefore (1) Given that DOH has been identified unequivocally as

the major and rapid metabolite of D and DA in rodents and humans, experiments are needed in the future to compare its *in vivo* efficacy with the parent “prodrug” compounds at PK-bioequivalent dosages to determine how much these pyranocoumarins contribute to anticancer activities of AGN. Similarly, the bioactivity comparisons should be expanded to other diseases/health conditions for which D or DA had been tested/implicated.

(2) Knowledge of DOH further metabolism should be obtained. DOH has been reported to exhibit high stability to oxidative and glucuronic metabolism in human and rat liver microsomes *in vitro* (Song *et al.*, 2011). The *in vivo* DOH metabolites should be investigated in rodents and blood samples from human intervention trials. Since CYP450 is implicated in DA and, to less extent, D hepatic conversion to DOH through oxygenated intermediates (Yoo *et al.*, 2007; Li *et al.*, 2013a) (Fig. 4), the extent and contribution of these upstream metabolic intermediates should not be overlooked, nor the impact and interactions with CYP450 inhibitor drugs.

(3) The PD targets of DOH and its further metabolites should be identified, in the case of anticancer bioactivity, in the malignant cells as well as cells that make up the tumor micro-environment (TME). The PD targets for many other health conditions and diseases should also be studied and re-visited. Our human PK data further motivate such endeavors because the likelihood for D and DA to directly bind proteins such as AR (Jiang *et al.*, 2006; Guo *et al.*, 2007) would be extremely low, if not nil. So far, Kang *et al.* (2008) identified a few DOH binding proteins from brain tissue by affinity chromatography and mass spectrometry. They include intracellular proteins heat shock protein 90kDa beta member 1 (GRP94 or HSP90B1), enolase 1 (ENO1), actin gamma 1 (ACTG1), heterogeneous nuclear ribonucleoprotein A2/B1 (HNRNPA2B1) and non-muscle myosin heavy polypeptide 9 (MYH9). The search strategy might be applicable to cancer cells, inflammatory and immune cells and other TME cell types.

(4) Besides pyranocoumarins, other components in AGN extract should be investigated. Whether these low abundant simple coumarins or furanocoumarins in AGN mediate or work in synergy with D/DA or DOH to exert anticancer and other bioactivities reviewed in this paper remain an open question.

An intriguing outcome of the superior efficacy of DOH-hybrid drugs against ischemia reperfusion stroke models than DOH or the counterpart drug each alone (Lee *et al.*, 2012; Yan *et al.*, 2013) warrant animal modeling to test whether the DOH-hybrid drug approach affords a brain targeting delivery across the BBB to enrich payload drugs in the hard-to-reach brain neurons. If affirmative, such an approach could be tested and leveraged to deliver cancer chemotherapy drugs to brain malignancies in addition to more potent drugs to prevent or treat stroke and other neurological disorders that could benefit from better cross BBB delivery.

Cell Culture “Mechanism” Studies

Because of the extensive PK and metabolism knowledge of pyranocoumarins (summarized in Fig. 4), the already published cell culture studies should be re-read through the critical lens of *in vivo* relevance. With exception of delivery routes that bypass the liver first

pass catabolism (e.g., *i.v.* infusion; transdermal; buccal/sublingual lozenge), the circulating levels of D and DA after AGN alcoholic extract supplement use would be expected to be extremely low when compared to that of DOH. While “mechanisms” reported so far as a result of exposure to double-to-triple digit micromolar concentrations of D or DA may suggest possible responses if such high local tissue concentrations can be achieved through special formulation, their probability of occurring *in vivo* is highly doubtful. More such “me-also” cell culture paper submissions to journals should be scrutinized by reviewers and editors for worthiness of publications, unless the authors provide convincing arguments for the likelihood of achievement of exposure range covered by the *in vitro* experiments. An example could be hepatocarcinoma cell culture modeling response to D and DA would be justified due to the expected high level of them moving from the portal vein to liver for first pass metabolism by hepatocyte CYP2C19 and 3A4 (Fig. 4).

In contrast, more studies in cell culture models should be designed with biological relevance and clear appreciation of the pyranocoumarin metabolism to study cellular and molecular responses for surrogate PD biomarkers to DOH or other AGN phytochemicals. In addition, exposure to D/DA and DOH combination based on their *in vivo* ratios should be compared to their exposure in isolation to explore possible interactions of the parent drugs and their metabolite. Such combination studies should also be expanded to include other non-pyranocoumarins.

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References

- Abd El-Aty, A.M., S.S. Shah, B.M. Kim, J.H. Choi, H.J. Cho, Y. Hee, B.J. Chang, H.C. Shin, K.B. Lee, M. Shimoda and J.H. Shim. *In vitro* inhibitory potential of decursin and decursinol angelate on the catalytic activity of cytochrome P-450 1A1/2, 2D15, and 3A12 isoforms in canine hepatic microsomes. *Arch. Pharm. Res.* 31: 1425–1435, 2008.
- Abdullayeva, R., M. Ganiyev and V. Asmetov. Effects of diumancal and decursinol on 5-hydroxytryptamine level in rat brain. *Georgian Med. News* 174: 84–88, 2009.
- Amat-Ur-Rasool, H., M. Ahmed, S. Hasnain and W.G. Carter. Anti-cholinesterase combination drug therapy as a potential treatment for Alzheimer's disease. *Brain Sci.* 11: 184, 2021.
- Bae, U.J., E.K. Choi, M.R. Oh, S.J. Jung, J. Park, T.S. Jung, T.S. Park, S.W. Chae and B.H. Park. *Angelica gigas* ameliorates hyperglycemia and hepatic steatosis in C57BL/KsJ-db/db mice via activation of AMP-activated protein kinase signaling pathway. *Am. J. Chin. Med.* 44: 1627–1638, 2016a.
- Bae, U.J., M.R. Oh, T.S. Jung, S.W. Chae and B.H. Park. Decursin and decursinol angelate-rich *Angelica gigas* Nakai extract suppresses de novo lipogenesis and alleviates nonalcoholic fatty liver disease and dyslipidemia in mice fed a high fat diet. *J. Funct. Foods* 31: 208–216, 2017.

- Bae, W.J., U.S. Ha, J.B. Choi, K.S. Kim, S.J. Kim, H.J. Cho, S.H. Hong, J.Y. Lee, Z. Wang, S.Y. Hwang and S.W. Kim. Protective effect of decursin extracted from *Angelica gigas* in male infertility via Nrf2/HO-1 signaling pathway. *Oxid. Med. Cell. Longev.* 2016: 5901098, 2016b.
- Bravo, P.L.W., H. Jin, H. Park, M.S. Kim, H. Matsui, H. Lee and J.W. Suh. Antithrombotic effect of the ethanol extract of *Angelica gigas* Nakai (AGE 232). *Life (Basel)* 11: 939, 2021.
- Chang, A., B.Y. Kwak, K. Yi and J.S. Kim. The effect of herbal extract (EstroG-100) on pre-, peri- and post-menopausal women: A randomized double-blind, placebo-controlled study. *Phytother. Res.* 26: 510–516, 2012.
- Chang, S.N., I. Khan, C.G. Kim, S.M. Park, D.K. Choi, H. Lee, B.S. Hwang, S.C. Kang and J.G. Park. Decursinol angelate arrest melanoma cell proliferation by initiating cell death and tumor shrinkage via induction of apoptosis. *Int. J. Mol. Sci.* 22: 4096, 2021.
- Choi, K.O., I. Lee, S.Y. Paik, D.E. Kim, J.D. Lim, W.S. Kang and S. Ko. Ultrafine *Angelica gigas* powder normalizes ovarian hormone levels and has antiosteoporosis properties in ovariectomized rats: Particle size effect. *J. Med. Food* 15: 863–872, 2012.
- Choi, S.S., K.J. Han, H.K. Lee, E.J. Han and H.W. Suh. Antinociceptive profiles of crude extract from roots of *Angelica gigas* Nakai in various pain models. *Biol. Pharm. Bull.* 26: 1283–1288, 2003a.
- Choi, S.S., K.J. Han, J.K. Lee, H.K. Lee, E.J. Han, D.H. Kim and H.W. Suh. Antinociceptive mechanisms of orally administered decursinol in the mouse. *Life Sci.* 73: 471–485, 2003b.
- Choi, Y.J., D.H. Kim, S.J. Kim, J. Kim, S.I. Jeong, C.H. Chung, K.Y. Yu and S.Y. Kim. Decursin attenuates hepatic fibrogenesis through interrupting TGF-beta-mediated NAD(P)H oxidase activation and Smad signaling *in vivo* and *in vitro*. *Life Sci.* 108: 94–103, 2014.
- Fontana, E., P.M. Dansette and S.M. Poli. Cytochrome p450 enzymes mechanism based inhibitors: Common sub-structures and reactivity. *Curr. Drug Metab.* 6: 413–454, 2005.
- Ge, Y., S.H. Yoon, H. Jang, J.H. Jeong and Y.M. Lee. Decursin promotes HIF-1alpha proteasomal degradation and immune responses in hypoxic tumour microenvironment. *Phytomedicine* 78: 153318, 2020.
- Guo, J., C. Jiang, Z. Wang, H.J. Lee, H. Hu, B. Malewicz, J.H. Lee, N.I. Baek, J.H. Jeong, D.K. Kim, K.S. Kang, S.H. Kim and J. Lu. A novel class of pyranocoumarin anti-androgen receptor signaling compounds. *Mol. Cancer Ther.* 6: 907–917, 2007.
- Guo, Y.R., H. Jin, M. Kim, M.B. Shin, J.H. Lee, S. Maeng, S.Y. Cha, J. Lee, Y.H. Koh, K.Y. Kim, S. Kang, H. Park and J.W. Suh. Synergistic neuroprotective effects of mature silkworm and *Angelica gigas* against scopolamine-induced mild cognitive impairment in mice and H2O2-induced cell death in HT22 mouse hippocampal neuronal cells. *J. Med. Food* 24: 505–516, 2021.
- Han, S.B., Y.H. Kim, C.W. Lee, S.M. Park, H.Y. Lee, K.S. Ahn, I.H. Kim and H.M. Kim. Characteristic immunostimulation by angelan isolated from *Angelica gigas* Nakai. *Immunopharmacology* 40: 39–48, 1998.
- Han, S.B., C.W. Lee, M.R. Kang, Y.D. Yoon, J.S. Kang, K.H. Lee, W.K. Yoon, S.K. Park and H.M. Kim. Pectic polysaccharide isolated from *Angelica gigas* Nakai inhibits melanoma cell metastasis and growth by directly preventing cell adhesion and activating host immune functions. *Cancer Lett.* 243: 264–273, 2006.
- He, L., Y. Pan, J. Yu, B. Wang, G. Dai and X. Ying. Decursin alleviates the aggravation of osteoarthritis via inhibiting PI3K-Akt and NF-kB signal pathway. *Int. Immunopharmacol.* 97: 107657, 2021.
- Hwang, J.T., S.H. Kim, H.J. Hur, H.J. Kim, J.H. Park, M.J. Sung, H.J. Yang, S.Y. Ryu, Y.S. Kim, M.R. Cha, M.S. Kim and D.Y. Kwon. Decursin, an active compound isolated from *Angelica gigas*, inhibits fat accumulation, reduces adipocytokine secretion and improves glucose tolerance in mice fed a high-fat diet. *Phytother. Res.* 26: 633–638, 2012.
- Jeon, Y.J., S.B. Han, K.S. Ahn and H.M. Kim. Activation of NF-kappaB/Rel in angelan-stimulated macrophages. *Immunopharmacology* 43: 1–9, 1999.

- Jeon, Y.J., S.B. Han, K.S. Ahn and H.M. Kim. Differential activation of murine macrophages by angelan and LPS. *Immunopharmacology* 49: 275–284, 2000.
- Jeon, Y.J., S.B. Han, S.H. Lee, H.C. Kim, K.S. Ahn and H.M. Kim. Activation of mitogen-activated protein kinase pathways by angelan in murine macrophages. *Int. Immunopharmacol.* 1: 237–245, 2001.
- Jeon, Y.J. and H.M. Kim. Experimental evidences and signal transduction pathways involved in the activation of NF-kappa B/Rel by angelan in murine macrophages. *Int. Immunopharmacol.* 1: 1331–1339, 2001.
- Ji, K.Y., D.H. Jung, B.J. Pyun, Y.J. Kim, J.Y. Lee, S. Choi, M.A. Jung, K.H. Song and T. Kim. *Angelica gigas* extract ameliorates allergic rhinitis in an ovalbumin-induced mouse model by inhibiting Th2 cell activation. *Phytomedicine* 93: 153789, 2021.
- Jiang, C., J. Guo, Z. Wang, B. Xiao, H.J. Lee, E.O. Lee, S.H. Kim and J. Lu. Decursin and decursinol angelate inhibit estrogen-stimulated and estrogen-independent growth and survival of breast cancer cells. *Breast Cancer Res.* 9: R77, 2007.
- Jiang, C., H.J. Lee, G.X. Li, J.M. Guo, B. Malewicz, Y. Zhao, E.O. Lee, J.H. Lee, M.S. Kim, S.H. Kim and J.X. Lu. Potent antiandrogen and androgen receptor activities of an *Angelica gigas*-containing herbal formulation: Identification of decursin as a novel and active compound with implications for prevention and treatment of prostate cancer. *Cancer Res.* 66: 453–463, 2006.
- Jin, B.R., M. Lee and H.J. An. Nodakenin represses obesity and its complications via the inhibition of the VLDLR signalling pathway *in vivo* and *in vitro*. *Cell Prolif.* 54: e13083, 2021.
- Joo, S.S., D. Park, S. Shin, J.H. Jeon, T.K. Kim, Y.J. Choi, S.H. Lee, J.S. Kim, S.K. Park, B.Y. Hwang, D.I. Lee and Y.B. Kim. Anti-allergic effects and mechanisms of action of the ethanolic extract of *Angelica gigas* in dinitrofluorobenzene-induced inflammation models. *Environ. Toxicol. Pharmacol.* 30: 127–133, 2010.
- Jung, D.J., A. Porzel and S. Huneck. Gigasol and other coumarins from *Angelica gigas*. *Phytochemistry* 30: 710–712, 1991.
- Jung, J.S., J.J. Yan and D.K. Song. Protective effect of decursinol on mouse models of sepsis: Enhancement of interleukin-10. *Korean J. Physiol. Pharmacol.* 12: 79–81, 2008.
- Jung, M.H., S.H. Lee, E.M. Ahn and Y.M. Lee. Decursin and decursinol angelate inhibit VEGF-induced angiogenesis via suppression of the VEGFR-2-signaling pathway. *Carcinogenesis* 30: 655–661, 2009.
- Jung, S.J., W.R. Kim, M.R. Oh, Y.S. Cha, B.H. Park and S.W. Chae. A randomized, double-blind, placebo-controlled clinical trial assessing the effects of *Angelica gigas* Nakai extract on blood triglycerides. *Nutrients* 12: 377, 2020.
- Kang, H.J., T.S. Yoon, D.G. Jeong, Y. Kim, J.W. Chung, J.S. Ha, S.S. Park, S.E. Ryu, S. Kim, K.H. Bae and S.J. Chung. Identification of proteins binding to decursinol by chemical proteomics. *J. Microbiol. Biotechnol.* 18: 1427–1430, 2008.
- Kang, S.Y. and Y.C. Kim. Neuroprotective coumarins from the root of *Angelica gigas*: Structure–activity relationships. *Arch. Pharm. Res.* 30: 1368–1373, 2007.
- Kang, S.Y., K.Y. Lee, K.A. Koo, J.S. Yoon, S.W. Lim, Y.C. Kim and S.H. Sung. ESP-102, a standardized combined extract of *Angelica gigas*, *Saururus chinensis* and *Schizandra chinensis*, significantly improved scopolamine-induced memory impairment in mice. *Life Sci.* 76: 1691–1705, 2005.
- Kang, S.Y., K.Y. Lee, M.J. Park, Y.C. Kim, G.J. Markelonis, T.H. Oh and Y.C. Kim. Decursin from *Angelica gigas* mitigates amnesia induced by scopolamine in mice. *Neurobiol. Learn. Mem.* 79: 11–18, 2003.
- Kang, S.Y., K.Y. Lee, S.H. Sung, M.J. Park and Y.C. Kim. Coumarins isolated from *Angelica gigas* inhibit acetylcholinesterase: Structure–activity relationships. *J. Nat. Prod.* 64: 683–685, 2001.
- Kim, D.H., W.Y. Jung, S.J. Park, J.M. Kim, S. Lee, Y.C. Kim and J.H. Ryu. Anti-amnesic effect of ESP-102 on A β (1–42)-induced memory impairment in mice. *Pharmacol. Biochem. Behav.* 97: 239–248, 2010.

- Kim, D.H., D.Y. Kim, Y.C. Kim, J.W. Jung, S. Lee, B.H. Yoon, J.H. Cheong, Y.S. Kim, S.S. Kang, K.H. Ko and J.H. Ryu. Nodakenin, a coumarin compound, ameliorates scopolamine-induced memory disruption in mice. *Life Sci.* 80: 1944–1950, 2007a.
- Kim, E.J., J. Kwon, S.H. Park, C. Park, Y.B. Seo, H.K. Shin, H.K. Kim, K.S. Lee, S.Y. Choi, D.H. Ryu and G.S. Hwang. Metabolite profiling of *Angelica gigas* from different geographical origins using (1)H NMR and UPLC-MS analyses. *J. Agric. Food. Chem.* 59: 8806–8815, 2011.
- Kim, H.M., J.S. Kang, S.K. Park, K. Lee, J.Y. Kim, Y.J. Kim, J.T. Hong, Y. Kim and S.B. Han. Antidiabetic activity of angelan isolated from *Angelica gigas* Nakai. *Arch. Pharm. Res.* 31: 1489–1496, 2008.
- Kim, J.H., S.K. Koh, H.J. Koh, Y.A. Kwon, S.H. Kim, J.G. Kim, T.E. Kim, J.W. Park, M.Y. Seo and Y.R. Song. A three month placebo-controlled clinical trial of INM 176 in the old aged subjects with memory impairment. *J. Korean Neuropsychiatric Assn.* 42: 254–262, 2003.
- Kim, J.Y., Y.D. Yoon, J.M. Ahn, J.S. Kang, S.K. Park, K. Lee, K. Bin Song, H.M. Kim and S.B. Han. Angelan isolated from *Angelica gigas* Nakai induces dendritic cell maturation through toll-like receptor 4. *Int. Immunopharmacol.* 7: 78–87, 2007b.
- Kim, K.J., J.T. Yeon, S.W. Choi, S.H. Moon, B.J. Ryu, R. Yu, S.J. Park, S.H. Kim and Y.J. Son. Decursin inhibits osteoclastogenesis by downregulating NFATc1 and blocking fusion of pre-osteoclasts. *Bone* 81: 208–216, 2015.
- Kim, K.M., M.J. Kim and J.S. Kang. Absorption, distribution, metabolism, and excretion of decursin and decursinol angelate from *Angelica gigas* Nakai. *J. Microbiol. Biotechnol.* 19: 1569–1572, 2009.
- Kim, M., M. Song, H.J. Oh, J. Hui, W. Bae, J. Shin, S.D. Ji, Y.H. Koh, J.W. Suh, H. Park and S. Maeng. Evaluating the memory enhancing effects of *Angelica gigas* in mouse models of mild cognitive impairments. *Nutrients* 12: 97, 2019.
- Kim, M.R., A.M. Abd El-Aty, J.H. Choi, K.B. Lee and J.H. Shim. Identification of volatile components in *Angelica* species using supercritical-CO₂ fluid extraction and solid phase microextraction coupled to gas chromatography-mass spectrometry. *Biomed. Chromatog. BMC* 20: 1267–1273, 2006a.
- Kim, M.R., A.M.A. Abd El-Aty, I.S. Kim and J.H. Shim. Determination of volatile flavor components in danggui cultivars by solvent free injection and hydrodistillation followed by gas chromatographic-mass spectrometric analysis. *J. Chromatogr. A* 1116: 259–264, 2006b.
- Kim, S.H., S.W. Lee, H.J. Park, S.H. Lee, W.K. Im, Y.D. Kim, K.H. Kim, S.J. Park, S. Hong and S.H. Jeon. Anti-cancer activity of *Angelica gigas* by increasing immune response and stimulating natural killer and natural killer T cells. *BMC Complement. Altern. Med.* 18: 218, 2018a.
- Kim, S.J., S.M. Ko, E.J. Choi, S.H. Ham, Y.D. Kwon, Y.B. Lee and H.Y. Cho. Simultaneous determination of decursin, decursinol angelate, nodakenin, and decursinol of *Angelica gigas* Nakai in human plasma by UHPLC-MS/MS: Application to pharmacokinetic study. *Molecules* 23: 2018b.
- Konoshima, M., H.J. Chi and K. Hata. Coumarins from the root of *Angelica gigas* Nakai. *Chem. Pharm. Bull. (Tokyo)* 16: 1139–1140, 1968.
- Lee, B., B. Sur, I. Shim, H. Lee and D.H. Hahm. *Angelica gigas* ameliorate depression-like symptoms in rats following chronic corticosterone injection. *BMC Complement. Altern. Med.* 15: 210, 2015.
- Lee, B., Z. Wu, S.H. Sung, T. Lee, K.S. Song, M.Y. Lee and K.H. Liu. Potential of decursin to inhibit the human cytochrome P450 2J2 isoform. *Food Chem. Toxicol.* 70: 94–99, 2014a.
- Lee, H.J., E.O. Lee, J.H. Lee, K.S. Lee, K.H. Kim, S.H. Kim and J. Lu. *In vivo* anti-cancer activity of Korean *Angelica gigas* and its major pyranocoumarin decursin. *Am. J. Chin. Med.* 37: 127–142, 2009.
- Lee, J.K., J.W. Jeong, T. Jang, G.W. Lee, H. Han, J.S. Kang and I.H. Kim. Decursin attenuates kainic acid-induced seizures in mice. *Neuroreport* 25: 1243–1249, 2014b.

- Lee, S., H. Jeon, S. Park, S. Lee, H.J. Chang and W. Eo. A potential treatment option for elderly non-Hodgkin lymphoma patients with multiple comorbidities: Two case reports and literature review. *Explore (NY)* 17: 265–269, 2020a.
- Lee, S., Y.S. Lee, S.H. Jung, K.H. Shin, B.K. Kim and S.S. Kang. Anti-tumor activities of decursinol angelate and decursin from *Angelica gigas*. *Arch. Pharm. Res.* 26: 727–730, 2003.
- Lee, S.E., J.H. Kim, C. Lim and S. Cho. Neuroprotective effect of *Angelica gigas* root in a mouse model of ischemic brain injury through MAPK signaling pathway regulation. *Chin. Med.* 15: 101, 2020b.
- Lee, S.E., C. Lim and S. Cho. *Angelica gigas* root ameliorates ischaemic stroke-induced brain injury in mice by activating the PI3K/AKT/mTOR and MAPK pathways. *Pharm. Biol.* 59: 662–671, 2021a.
- Lee, S.W., H.J. Park, S.H. Kim, S. Shin, K.H. Kim, S.J. Park, S. Hong and S.H. Jeon. TLR4-dependent effects of ISAg treatment on conventional T cell polarization *in vivo*. *Anim. Cells Syst. (Seoul)* 23: 184–191, 2019a.
- Lee, T.H., J.H. Park, J.D. Kim, J.C. Lee, I.H. Kim, Y. Yim, S.K. Lee, B.C. Yan, J.H. Ahn, C.H. Lee, K.Y. Yoo, J.H. Choi, I.K. Hwang, J.H. Park and M.H. Won. Protective effects of a novel synthetic alpha-lipoic acid-decursinol hybrid compound in experimentally induced transient cerebral ischemia. *Cell. Mol. Neurobiol.* 32: 1209–1221, 2012.
- Lee, T.K., I.J. Kang, H. Sim, J.C. Lee, J.H. Ahn, D.W. Kim, J.H. Park, C.H. Lee, J.D. Kim, M.H. Won and S.Y. Choi. Therapeutic effects of decursin and *Angelica gigas* Nakai root extract in gerbil brain after transient ischemia via protecting BBB leakage and astrocyte endfeet damage. *Molecules* 26: 2161, 2021b.
- Lee, W., O. Yuseok, C. Lee, S.Y. Jeong, J.H. Lee, M.C. Baek, G.Y. Song and J.S. Bae. Suppressive activities of KC1-3 on HMGB1-mediated septic responses. *Biochem. Pharmacol.* 163: 260–268, 2019b.
- Lee, W., O. Yuseok, S. Yang, B.S. Lee, J.H. Lee, E.K. Park, M.C. Baek, G.Y. Song and J.S. Bae. JH-4 reduces HMGB1-mediated septic responses and improves survival rate in septic mice. *J. Cell. Biochem.* 120: 6277–6289, 2019c.
- Li, L., J. Zhang, A.A. Shaik, Y. Zhang, L. Wang, C. Xing, S.H. Kim and J. Lu. Quantitative determination of decursin, decursinol angelate, and decursinol in mouse plasma and tumor tissue using liquid–liquid extraction and HPLC. *Planta Med.* 78: 252–259, 2012.
- Li, L., J. Zhang, C. Xing, S.H. Kim, C. Jiang and J. Lu. *In vitro* metabolism of pyranocoumarin isomers decursin and decursinol angelate by liver microsomes from man and rodents. *Planta Med.* 79: 1536–1544, 2013a.
- Li, L., J. Zhang, C. Xing, S.H. Kim and J. Lu. Single oral dose pharmacokinetics of decursin, decursinol angelate, and decursinol in Rats. *Planta Med.* 79: 275–280, 2013b.
- Lu, J., J. Zhang, L. Li, C. Jiang and C. Xing. Cancer chemoprevention with Korean *Angelica*: Active compounds, pharmacokinetics, and human translational considerations. *Curr. Pharmacol. Rep.* 1: 373–381, 2015.
- Mahat, B., J.W. Chae, I.H. Baek, G.Y. Song, J.S. Song, J.Y. Ma and K.I. Kwon. Biopharmaceutical characterization of decursin and their derivatives for drug discovery. *Drug. Dev. Ind. Pharm.* 39: 1523–1530, 2013.
- Moon, H.I., J.H. Lee, Y.C. Lee and K.S. Kim. Antiplasmodial and cytotoxic activity of coumarin derivatives from dried roots of *Angelica gigas* Nakai *in vitro*. *Immunopharmacol. Immunotoxicol.* 33: 663–666, 2011.
- Oh, S.R., S. Ok, T.S. Jung, S.O. Jeon, J.M. Park, J.W. Jung and D.S. Ryu. Protective effect of decursin and decursinol angelate-rich *Angelica gigas* Nakai extract on dextran sulfate sodium-induced murine ulcerative colitis. *Asian Pac. J. Trop. Med.* 10: 864–870, 2017.
- Oh, T.W., K.H. Park, H.W. Jung and Y.K. Park. Neuroprotective effect of the hairy root extract of *Angelica gigas* Nakai on transient focal cerebral ischemia in rats through the regulation of angiogenesis. *BMC Complement. Altern. Med.* 15: 101, 2015.

- Pak, S., B. Thapa and K. Lee. Decursinol angelate mitigates sepsis induced by methicillin-resistant staphylococcus aureus infection by modulating the inflammatory responses of macrophages. *Int. J. Mol. Sci.* 22: 10950, 2021.
- Park, H.S., B. Kim, J.H. Oh, Y.C. Kim and Y.J. Lee. First-pass metabolism of decursin, a bioactive compound of *Angelica gigas*, in rats. *Planta Med.* 78: 909–913, 2012a.
- Park, H.Y., S.B. Kwon, N.K. Heo, W.J. Chun, M.J. Kim and Y.S. Kwon. Constituents of the stem of *Angelica gigas* with rat lens aldose reductase inhibitory activity. *J. Korean Soc. Appl. Biol. Chem.* 54: 194–199, 2011.
- Park, I.S., B. Kim, Y. Han, H. Yang, U. Cho, S.I. Kim, J.H. Kim, J.H. Yoon Park, K.W. Lee and Y.S. Song. Decursin and decursinol angelate suppress adipogenesis through activation of beta-catenin signaling pathway in human visceral adipose-derived stem cells. *Nutrients* 12: 13, 2019.
- Park, J.H., N.I. Park, H. Xu and S.U. Park. Cloning and characterization of phenylalanine ammonia-lyase and cinnamate 4-hydroxylase and pyranocoumarin biosynthesis in *Angelica gigas*. *J. Nat. Prod.* 73: 1394–1397, 2010.
- Park, R.J., N.J. Kim, K.T. Lee and S.H. Seo. Comparative studies on concentration of decursinol in plasma after oral administration of *Angelicae gigantis radix* extract and combined use of decursin and *Cnidii rhizoma* extract or *Bupleuri radix* extract in rats. *Korean J. Pharmacognosy* 32: 72–78, 2001.
- Park, S.J., H.J. Jung, M.S. Son, J.M. Jung, D.H. Kim, I.H. Jung, Y.B. Cho, E.H. Lee and J.H. Ryu. Neuroprotective effects of INM-176 against lipopolysaccharide-induced neuronal injury. *Pharmacol. Biochem. Behav.* 101: 427–433, 2012b.
- Park, S.J., J.M. Jung, H.E. Lee, Y.W. Lee, D.H. Kim, J.M. Kim, J.G. Hong, C.H. Lee, I.H. Jung, Y.B. Cho, D.S. Jang and J.H. Ryu. The memory ameliorating effects of INM-176, an ethanolic extract of *Angelica gigas*, against scopolamine- or Abeta(1–42)-induced cognitive dysfunction in mice. *J. Ethnopharmacol.* 143: 611–620, 2012c.
- Pathak, M.A. and T.B. Fitzpatrick. The evolution of photochemotherapy with psoralens and UVA (PUVA): 2000 BC to 1992 AD. *J. Photochem. Photobiol. B* 14: 3–22, 1992.
- Piao, J., J.Y. Lee, J.B. Weon, C.J. Ma, H.J. Ko, D.D. Kim, W.S. Kang and H.J. Cho. *Angelica gigas* Nakai and Soluplus-based solid formulations prepared by hot-melting extrusion: Oral absorption enhancing and memory ameliorating effects. *PLoS One* 10: e0124447, 2015.
- Piao, X.L., J.H. Park, H. Cui, D.H. Kim and H.H. Yoo. Development of gas chromatographic mass spectrometry-pattern recognition method for the quality control of Korean *Angelica*. *J. Pharm. Biomed. Anal.* 44: 1163–1167, 2007.
- Reddy, V.L.N., A.N. Jadhav, B. Avula and I.A. Khan. Isolation of pyranocoumarins from *Angelica gigas*. *Nat. Prod. Commun.* 3: 785–790, 2008.
- Seo, Y.J., M.S. Kwon, S.H. Park, Y.B. Sim, S.M. Choi, G.H. Huh, J.K. Lee and H.W. Suh. The analgesic effect of decursinol. *Arch. Pharm. Res.* 32: 937–943, 2009.
- Sharifipour, M., E. Izadpanah, B. Nikkhoo, S. Zare, A. Abdolmaleki, K. Hassanzadeh, F. Moradi and K. Hassanzadeh. A new pharmacological role for donepezil: Attenuation of morphine-induced tolerance and apoptosis in rat central nervous system. *J. Biomed. Sci.* 21: 6, 2014.
- Shin, S., J.H. Jeon, D. Park, J.Y. Jang, S.S. Joo, B.Y. Hwang, S.Y. Choe and Y.B. Kim. Anti-inflammatory effects of an ethanol extract of *Angelica gigas* in a carrageenan-air pouch inflammation model. *Exp. Anim.* 58: 431–436, 2009.
- Son, D.B., W. Choi, M. Kim, E.J. Go, D. Jeong, C.K. Park, Y.H. Kim, H. Lee and J.W. Suh. Decursin alleviates mechanical allodynia in a paclitaxel-induced neuropathic pain mouse model. *Cells* 10: 547, 2021.
- Son, S.H., K.K. Park, S.K. Park, Y.C. Kim, Y.S. Kim, S.K. Lee and W.Y. Chung. Decursin and decursinol from *Angelica gigas* inhibit the lung metastasis of murine colon carcinoma. *Phytother. Res.* 25: 959–964, 2011.

- Song, J.S., J.W. Chae, K.R. Lee, B.H. Lee, E.J. Choi, S.H. Ahn, K.I. Kwon and M.A. Bae. Pharmacokinetic characterization of decursinol derived from *Angelica gigas* Nakai in rats. *Xenobiotica* 41: 895–902, 2011.
- Tang, S.N., P. Jiang, S. Kim, J. Zhang, C. Jiang and J. Lu. Interception targets of *Angelica gigas* Nakai root extract vs. pyranocoumarins in prostate early lesions and neuroendocrine carcinomas in TRAMP mice. *Cancer Prev. Res. (Phila)* 14: 635–648, 2021.
- Tang, S.N., J. Zhang, W. Wu, P. Jiang, M. Puppala, Y. Zhang, C. Xing, S.H. Kim, C. Jiang and J. Lu. Chemopreventive effects of Korean *Angelica* versus its major pyranocoumarins on two lineages of transgenic adenocarcinoma of mouse prostate carcinogenesis. *Cancer Prev. Res. (Phila)* 8: 835–844, 2015.
- Thapa, B., S. Pak, H.J. Kwon and K. Lee. Decursinol angelate ameliorates dextran sodium sulfate-induced colitis by modulating type 17 helper T cell responses. *Biomol. Ther. (Seoul)* 27: 466–473, 2019.
- Wang, X., T. Zheng, J.H. Kang, H. Li, H. Cho, R. Jeon, J.H. Ryu and M. Yim. Decursin from *Angelica gigas* suppresses RANKL-induced osteoclast formation and bone loss. *Eur. J. Pharmacol.* 774: 34–42, 2016.
- Woo, J.H., T.W. Ha, J.S. Kang, J.T. Hong and K.W. Oh. Potentiation of decursinol angelate on pentobarbital-induced sleeping behaviors via the activation of GABAA-ergic systems in rodents. *Korean J. Physiol. Pharmacol.* 21: 27–36, 2017.
- Wu, W., S.N. Tang, Y. Zhang, M. Puppala, T.K. Cooper, C. Xing, C. Jiang and J. Lu. Prostate cancer xenograft inhibitory activity and pharmacokinetics of decursinol, a metabolite of *Angelica gigas* pyranocoumarins, in mouse models. *Am. J. Chin. Med.* 45: 1773–1792, 2017.
- Yan, B.C., J.H. Park, B.N. Shin, J.H. Ahn, I.H. Kim, J.C. Lee, K.Y. Yoo, I.K. Hwang, J.H. Choi, J.H. Park, Y.L. Lee, H.W. Suh, J.G. Jun, Y.G. Kwon, Y.M. Kim, S.H. Kwon, S. Her, J.S. Kim, B.H. Hyun, C.K. Kim, J.H. Cho, C.H. Lee and M.H. Won. Neuroprotective effect of a new synthetic aspirin-decursinol adduct in experimental animal models of ischemic stroke. *PLoS One* 8: e74886, 2013.
- Yan, F., P.F. Thall, K.H. Lu, M.R. Gilbert and Y. Yuan. Phase I-II clinical trial design: A state-of-the-art paradigm for dose finding. *Ann. Oncol.* 29: 694–699, 2018.
- Yan, J.J., D.H. Kim, Y.S. Moon, J.S. Jung, E.M. Ahn, N.I. Baek and D.K. Song. Protection against beta-amyloid peptide-induced memory impairment with long-term administration of extract of *Angelica gigas* or decursinol in mice. *Prog. Neuropsychopharmacol. Biol. Psychiatry* 28: 25–30, 2004.
- Yim, D., R.P. Singh, C. Agarwal, S. Lee, H. Chi and R. Agarwal. A novel anticancer agent, decursin, induces G1 arrest and apoptosis in human prostate carcinoma cells. *Cancer Res.* 65: 1035–1044, 2005.
- Yoo, H.H., M.W. Lee, Y.C. Kim, C.H. Yun and D.H. Kim. Mechanism-based inactivation of cytochrome p450 2A6 by decursinol angelate isolated from *Angelica gigas*. *Drug Metab. Dispos.* 35: 1759–1765, 2007.
- Yoo, J.S., E.M. Ahn, M.C. Song, M.H. Bang, D.H. Kim, M.W. Han, H.Y. Kwak, D.Y. Lee, H.N. Lyu and N.I. Baek. Quantitative analysis of coumarins from *Angelica gigas* using H-1-NMR. *Food Sci. Biotechnol.* 17: 573–577, 2008.
- Zhang, J., L. Li, T.W. Hale, W. Chee, C. Xing, C. Jiang and J. Lu. Single oral dose pharmacokinetics of decursin and decursinol angelate in healthy adult men and women. *PLoS One* 10: e0114992, 2015a.
- Zhang, J., L. Li, C. Jiang, C. Xing, S.H. Kim and J. Lu. Anti-cancer and other bioactivities of Korean *Angelica gigas* Nakai (AGN) and its major pyranocoumarin compounds. *Anticancer Agents Med. Chem.* 12: 1239–1254, 2012a.
- Zhang, J., L. Li, S. Tang, T.W. Hale, C. Xing, C. Jiang and J. Lu. Cytochrome P450 isoforms in the metabolism of decursin and decursinol angelate from Korean *Angelica*. *Am. J. Chin. Med.* 43: 1211–1230, 2015b.

- Zhang, J., L. Wang, Y. Zhang, L. Li, S. Tang, C. Xing, S.H. Kim, C. Jiang and J. Lu. Chemopreventive effect of Korean Angelica root extract on TRAMP carcinogenesis and integrative “omic” profiling of affected neuroendocrine carcinomas. *Mol. Carcinog.* 54: 1567–1583, 2015c.
- Zhang, Y., A.A. Shaik, C. Xing, Y. Chai, L. Li, J. Zhang, W. Zhang, S.H. Kim, J. Lu and C. Jiang. A synthetic decursin analog with increased *in vivo* stability suppresses androgen receptor signaling *in vitro* and *in vivo*. *Invest. New Drugs* 30: 1820–1829, 2012b.
- Zhao, K.J., T.T. Dong, P.F. Tu, Z.H. Song, C.K. Lo and K.W. Tsim. Molecular genetic and chemical assessment of radix *Angelica* (Danggui) in China. *J. Agric. Food. Chem.* 51: 2576–2583, 2003.